

Ammonia-oxidizing bacteria and archaea exhibit differential nitrogen source preferences

Wei Qin^{1, 2†*}, Stephany P. Wei^{2†}, Yue Zheng^{3†}, Eunkyung Choi^{4†}, Xiangpeng Li¹, Juliet Johnston⁵, Xianhui Wan⁶, Britt Abrahamson², Zachary Flinkstrom², Baozhan Wang⁷, Hanyan Li¹, Lei Hou^{1,3}, Qing Tao¹, Wyatt, W. Chlouber¹, Xin Sun⁸, Michael Wells¹, Long Ngo¹, Kristopher A. Hunt², Hidetoshi Urakawa⁹, Xuanyu Tao¹, Dongyu Wang¹, Xiaoyuan Yan¹⁰, Dazhi Wang³, Chongle Pan¹, Peter K. Weber⁵, Jiandong Jiang⁷, Jizhong Zhou¹, Yao Zhang³, David A. Stahl², Bess B. Ward⁶, Xavier Mayali⁵, Willm Martens-Habbena^{4*}, Mari -Karoliina H. Winkler^{2*}

¹School of Biological Sciences, Institute for Environmental Genomics, University of Oklahoma, Norman, Oklahoma, USA.

²Department of Civil and Environmental Engineering, University of Washington, Seattle, Washington, USA.

³State Key Laboratory of Marine Environmental Science, Xiamen University, Xiamen, Fujian, China.

⁴Department of Microbiology and Cell Science, University of Florida, Davie, Florida, USA.

⁵Lawrence Livermore National Laboratory, Livermore, California, USA.

⁶Department of Geosciences, Princeton University, Princeton, New Jersey, USA.

⁷College of Life Sciences, Nanjing Agricultural University, Nanjing, Jiangsu, China.

⁸Department of Ecology and Evolutionary Biology, Yale University, New Haven, Connecticut, USA.

⁹Department of Ecology and Environmental Studies, Florida Gulf Coast University, Fort Myers, Florida, USA.

¹⁰State Key Laboratory of Soil and Sustainable Agriculture, Institute of Soil Science, Chinese Academy of Sciences, Nanjing, China.

[†]These authors contributed equally to this work: Wei Qin, Stephany P. Wei, Yue Zheng, Eunkyung Choi.

^{*}To whom correspondence should be addressed: Wei Qin (weiqin@ou.edu), Willm Martens-Habbena (w.martenshabbena@ufl.edu), Mari Winkler (mwinkler@uw.edu).

Abstract

Ammonia-oxidizing microorganisms (AOM) contribute to one of the largest nitrogen fluxes in the global nitrogen budget. Four distinct lineages of AOM, ammonia-oxidizing archaea (AOA), ammonia-oxidizing bacteria (AOB), and complete ammonia oxidizers (comammox), are thought to compete for ammonia as their primary nitrogen substrate. Additionally, many AOM species can utilize urea as an alternative energy and nitrogen source through hydrolysis to ammonia. How the coordination of ammonia and urea metabolism in AOM influences their ecology remains poorly understood. Here, we use stable isotope tracing, kinetics, and transcriptomics experiments to show that representatives of the AOM lineages employ distinct regulatory strategies for ammonia or urea utilization, thereby minimizing direct substrate competition. The tested AOA and comammox species preferentially used ammonia over urea, while beta-proteobacterial AOB favored urea utilization, repressed ammonia transport in the presence of urea, and showed higher affinity for urea than ammonia. Characterized gamma-proteobacterial AOB co-utilized both substrates. These results reveal contrasting niche adaptation and coexistence patterns among the major AOM lineages.

Introduction

Ammonia-oxidizing microorganisms (AOM) account for the oxidation of ~2,330 Tg N per year, representing one of the largest nitrogen fluxes in the global nitrogen budget¹. Ammonia-oxidizing bacteria and archaea (AOB, AOA) oxidize ammonia (hereafter considered as total ammonia, $\text{NH}_3 + \text{NH}_4^+$) to nitrite^{2,3}, which is then oxidized to nitrate by nitrite-oxidizing bacteria (NOB), comprising the two-step process of nitrification. Recently, complete ammonia oxidizers (comammox), capable of oxidizing ammonia to nitrate, were discovered within the NOB genus *Nitrospira*^{4,5}. Given that all AOM use ammonia as energy and nitrogen (N) sources for growth, their specific affinity for ammonia has been considered the main determinant of their competition and niche partitioning⁶⁻⁸. However, fundamental questions remain about the biochemical and biophysical controls of their specific affinities for ammonia^{9,10}. Although different lineages of AOM are thought to compete for their primary N substrate, ammonia, they often grow concomitantly in soil, freshwater, estuarine, and subsurface ecosystems, raising the question of other adaptative strategies that allow for their co-occurrence.

Microorganisms adapt to changing resource availability by either utilizing multiple substrates simultaneously or preferentially using those that support optimal growth with least investment in enzyme inventory, the later exemplified by the glucose-lactose diauxie in *Escherichia coli*¹¹. Most studies have focused on how microorganisms metabolize carbon (C) mixtures¹² or mixed N sources for anabolism^{13,14}. Less is known about selection of alternative N substrates for catabolic energy conversion, a capacity unique to nitrifying microorganisms¹. For example, an adaptive trait of many AOM species is the capacity to utilize urea as an energy and N source through hydrolysis to ammonia^{4,15-20}.

Urea is an important and widely available organic nitrogen compound in soil and aquatic environments, released as nitrogenous waste from both eukaryotes and prokaryotes and subsequently hydrolyzed by ureolytic microorganisms^{21,22}. Urea-based fertilizers are a mainstay

in modern agriculture, and the intensive use of urea in fertilizers has greatly increased urea export to surrounding freshwater and coastal ecosystems. Since the urease enzyme does not require a high-energy cofactor, such as ATP²³, recent studies in marine systems suggested that ammonia and urea may be co-utilized for energy and N requirements²⁴. However, utilization or even co-utilization of urea would require an investment in the synthesis of the urea transporter, urease, and accessory proteins, a significant cost for autotrophic AOM using the low energy-yielding ammonia oxidation process. Thus, there is a compelling need for better mechanistic understanding of urea utilization by ammonia oxidizers having capacity to use this substrate.

Here we use a combination of stable isotope tracing, kinetics, and transcriptomics studies to resolve the distinct regulatory strategies for selective uptake and metabolism of ammonia and urea by AOA, beta- and gamma-proteobacterial AOB, and comammox. By revealing that ammonia oxidation and assimilation are both transport-dependent, those analyses also provided insights into how reduced N is partitioned between energy generation and biosynthesis, and mechanistic understanding of previously unrecognized adaptive regulation of cellular affinity for ammonia. Together these findings advance understanding of niche separation and resource partitioning among lineages of AOM and should foster improved nitrogen cycle models.

Results

AOM lineages exhibit distinct N preferences for catabolism

Comparative analysis of available AOM genomes showed that more than 50%, 60%, and 80% of AOA, AOB, and comammox species, respectively, contain genes encoding urea permease, urease, and accessory proteins (Extended Data Fig. 1). Thus, the potential for urea uptake and utilization is widespread among AOM. Seven phylogenetically and ecologically diverse AOM species, including a marine AOA (*Nitrosopumilus piranensis* D3C)¹⁵, a soil AOA (*Nitrososphaera viennensis* EN76)¹⁶, three β -AOB (*Nitrosospira lacus* APG3, *Nitrosospira multiformis* ATCC 25196, and *Nitrosomonas ureae* Nm10)^{17,19,20}, a marine γ -AOB

(*Nitrosococcus oceani* ATCC 19707)¹⁸, and a comammox (*Nitrospira inopinata* ENR4)⁴ were selected for characterization of N substrate preference (Extended Data Fig. 2). As expected, all strains grew by stoichiometric conversion of ammonia or urea to nitrite or nitrate when grown in media containing a single N substrate (Fig. 1). No extracellular urease activity was observed for any of the strains (Supplementary Fig. 1). When grown on mixtures of equal amounts of ammonia- and urea-N, three distinct N utilization patterns emerged (Fig. 1): While the tested AOA and comammox species depleted ammonia before hydrolyzing significant amounts of urea (Fig. 1a–c), β -AOB species used urea before consuming substantial amounts of free ammonia (Fig. 1d–f). In contrast, γ -AOB species simultaneously consumed ammonia and urea (Fig. 1g).

Notably, the marine AOA *N. piranensis* exhibited a marked diauxic lag between the two growth phases (Fig. 1a), reflecting a longer adaptive transition from ammonia to urea consumption. The diauxic lag phase was less pronounced for the soil AOA *N. viennensis*, displaying a rapid transition from ammonia to urea consumption (Fig. 1b). The comammox *N. inopinata* also displayed preferential consumption of ammonia when grown in the mixed N species medium (Fig. 1c). However, *N. inopinata* started to consume urea prior to the complete exhaustion of ammonia, suggesting a less stringent diauxic transition (Fig. 1c, Supplementary Discussion).

During the period of urea consumption, free ammonia concentrations increased in β -AOB *N. lacus* and *N. multiformis* (Fig. 1d and e) and decreased slightly in *N. ureae* cultures (Fig. 1f). Only after near complete consumption of urea did ammonia concentrations decrease substantially (Fig. 1d–f). The same results were observed irrespective of whether cells were pre-grown on either ammonia or urea. Unlike all other tested strains, the γ -AOB *N. oceani* co-utilized ammonia and urea without apparent preference, but only growing on urea at a rate comparable to growth on ammonia in the presence of excess ammonia in the medium (Supplementary Fig. 2). Without ammonia supplementation, *N. oceani* required almost a year to consume 500 μ M urea (Fig. 1g and Supplementary Fig. 3), corresponding to an extended average generation time of 23 days.

Consistent with the preferential use of ammonia, characterized AOA and comammox species showed significantly higher specific growth rates ($P < 0.05$) on ammonia, compared to urea after ammonia exhaustion on a mixture of ammonia and urea (Fig. 1a–c). In contrast, the specific growth rates of the β -AOB strains on urea were comparable or even higher ($P < 0.05$) than on ammonia in the mixed medium (Fig. 1d–f), indicating their high urea uptake and hydrolysis activities. Together, these data indicated three contrasting N source preferences and regulation patterns among β -AOB, AOA and comammox, and γ -AOB. Whereas the tested AOA and comammox species repressed the urea-utilizing functions when ammonia was available, β -AOB activated urea transport and hydrolysis rapidly upon exposure to urea.

Although the *Nitrosospira* β -AOB strains preferentially consumed urea, release of free ammonia was observed when grown on urea or a urea-ammonia mixture (Fig. 1d and e). To evaluate the use of free ammonia in the presence of urea, and to investigate the fate of alternative N sources, we used ^{15}N isotope labeling to track conversion of ammonia or urea-N throughout batch culture growth of *N. lacus*, which exhibited the most significant release of free ammonia when grown in the mixed medium (Fig. 1d). *N. lacus* was grown on a mixture of ammonia and urea, with one or the other N substrate labeled with ^{15}N . For the $^{15}\text{NH}_3$ and ^{15}N -urea tracers, $^{15}\text{NO}_2^-$ production was measured (Fig. 2), and for the ^{15}N -urea tracer, $^{15}\text{NH}_3$ released to the medium was also measured (Extended Data Fig. 3, Supplementary Discussion). The temporal distribution of extracellular N isotopes exhibited three distinct phases (Fig. 2). The small amount of $^{15}\text{NO}_2^-$ produced during lag phase (0–50 h) originated mostly from $^{15}\text{NH}_3$ (Fig. 2a), reflecting metabolism of the ammonia-grown inoculum culture. However, that small amount of $^{15}\text{NO}_2^-$ was rapidly diluted by unlabeled nitrite originating from urea throughout exponential growth (~100–150 h), reflecting a sharp drop in $^{15}\text{NH}_3$ contribution to $^{15}\text{NO}_2^-$ production (Fig. 2a). Conversely, ^{15}N -urea labeling showed that NO_2^- almost exclusively originated from urea in this phase (Fig. 2b). In the presence of ^{15}N -urea, $^{15}\text{NH}_3$ was continuously released from cells until urea depletion (Extended Data Fig. 3b). During the third phase following depletion of urea

around 150 h, ammonia consumption rapidly resumed, and all nitrite then originated from ammonia until the cessation of growth (Fig. 2a). Collectively, these results provided unequivocal evidence that only the ammonia released from urea hydrolysis in the cytoplasm was oxidized and that the oxidation of extracellular ammonia was repressed in the presence of urea.

Nitrogen of preferred substrate is also used for anabolism

Since ammonia- and urea-N are assimilated by AOM, we also examined their incorporation in representative AOM species at single-cell resolution using stable isotope labeling in combination with NanoSIMS. Following growth on dual-labeled $^{13}\text{C}^{15}\text{N}$ -urea (plus unlabeled NH_3 and bicarbonate) or $^{15}\text{NH}_3$ and ^{13}C -bicarbonate (plus unlabeled urea) (Extended Data Figs. 4–7, Supplementary Fig. 4), cells for NanoSIMS were collected during consumption of the preferred N substrate (T1) and after depletion of all N substrates (T2) (Extended Data Fig. 4a). *N. oceani*, oxidizing both substrates simultaneously, was harvested at late exponential phase (T1) and after both N substrates were exhausted (T2). Urea-derived C was shown to be only a minor carbon source (Extended Data Fig. 7, Supplementary Discussion).

Although *N. lacus* released significant ammonia into the medium when using urea, the extracellular ammonia pool was not appreciably assimilated during urea consumption. Results from T1 showed $83.9 \pm 6.9\%$ new N incorporation from urea and $16.1 \pm 1.4\%$ from ammonia (Extended Data Fig. 4b, c), indicating that urea-N was preferentially assimilated in the first phase of growth. The assimilation of ammonia during lag phase reflected a period of acclimation of the ammonia-preadapted *N. lacus* to growth on urea-ammonia. At T2, new N incorporation from urea had decreased to $54.2 \pm 2.5\%$ and that derived from ammonia had increased to $45.8 \pm 1.9\%$ (Extended Data Fig. 4b, c). Thus, even in the presence of free ammonia, urea-derived N was preferentially assimilated by *N. lacus* until urea depletion. A similar set of labeling experiments was performed with the marine AOA *N. piranensis*, soil AOA *N. viennensis*, and comammox *N. inopinata*. In contrast to *N. lacus*, they all preferentially assimilated N from ammonia (Extended

Data Fig. 4b, c, Supplementary Discussion). In addition, γ -AOB *N. oceanii* co-assimilated ammonia and urea.

Kinetic characterization of ammonia and urea oxidation

To determine whether N preference was associated with different cellular kinetics, we characterized ammonia- and urea-dependent oxidation kinetics using microrespirometry (Supplementary Table 2). Ammonia and urea-dependent oxygen consumption followed Michaelis-Menten kinetics in all AOM investigated here (Extended Data Fig. 8). Whereas the marine AOA *N. piranensis* showed similar apparent half-saturation constants ($K_{m(app)}$) for ammonia and urea (0.70 ± 0.22 and 1.65 ± 0.82 μ M, respectively), the soil AOA *N. viennensis* had comparable $K_{m(app)}$ for ammonia (0.68 ± 0.16 μ M), but significantly higher $K_{m(app)}$ for urea (8.97 ± 1.27 μ M) (Fig. 3a). Correspondingly, the specific affinities (a°) for ammonia and urea in *N. piranensis* and ammonia in *N. viennensis* were similar, whereas *N. viennensis* a° for urea was approximately 10-fold lower (Fig. 3b). The comammox *N. inopinata* showed comparably high affinity for ammonia as AOA species. However, unlike the AOA species, *N. inopinata* also showed a very high a° for urea, which exceeds that of *N. piranensis* and *N. viennensis* by more than 20- and even 90-fold, respectively (Fig. 3b, Supplementary Table 2). *N. inopinata* contains an ABC-type urea transporter (Extended Data Fig. 2) that may support its high affinity for urea via ATP-dependent urea uptake. Notably, the β -AOB *N. lacus* and *N. multiformis* had 2.7–6.2-fold higher a° for urea than for ammonia (Fig. 3b, Supplementary Table 2), comparable to or even higher ($P < 0.05$) than that of *N. viennensis* (Fig. 3b), suggesting some AOB common in soils and freshwater sediments might effectively compete with AOA for urea.

We further tested ammonia affinities of urea-grown strains. To our surprise, only *N. viennensis* showed similar $K_{m(app)}$ and a° independent of prior growth on ammonia or urea, whereas affinity for ammonia was significantly higher ($P < 0.05$) in urea-grown cells of *N. piranensis* and *N. inopinata* than ammonia-grown cells (Fig. 3). The capacity of marine AOA and comammox to

alter their uptake kinetics for ammonia when growing on urea suggests an adaptation for more efficient uptake of their primary N source. Taken together, the kinetic characteristics of AOM species track their N substrate preferences. *N. inopinata* is a notable exception. Its unusually high urea affinity through ATP-dependent transport might provide competitive advantage in environments where both N sources are available.

Transcription linked to growth on single and mixed N sources

Growth kinetics and gene transcription suggested a regulatory response to the availability of different N sources, as observed for marine AOA *N. piranensis* and soil AOA *N. viennensis* (Fig. 4a, b). Cells were characterized at mid-exponential phase on a single N source (T1) and at 24 hours after introduction of the alternative N substrate (T2) (Fig. 4a, b). Additional samples were taken during the extended lag in *N. piranensis* growth during its transition from ammonia to urea (AU-T3) and after its consumption of the newly supplemented N species (AU-T4 & UA-T3) (Fig. 4a).

Among characterized AOA species, ammonia is used as a source of both energy and electrons through its oxidation by AMO, and for biosynthesis through the activities of glutamate dehydrogenase (GDH) and glutamine synthetase (GS)^{15,25}. Transcription of *amoA* (AMO alpha subunit), *gdh*, and *gs* was relatively unchanged during exponential growth of *N. piranensis* and *N. viennensis* on either N substrate (AU-T1 vs. UA-T1, Supplementary Table 3). In contrast, the transcription of *amt1* (putative high-affinity ammonium transporter), *ut* (urea transporter), *ure* (urease), and *glnB* (P_{II} protein) were highly responsive to the addition of the alternative N sources (Fig. 4a, b).

Initial low transcription of *ure* and DUR3-type *ut* (NPIRD3C_1395) in ammonia-grown *N. piranensis* increased to very high levels after switching to urea consumption (Fig. 4a). In turn, *ut* transcription was significantly depressed upon addition of ammonia to cultures growing on urea (UA-T2 vs. T1, $P = 1.01 \times 10^{-13}$), dropping to its lowest levels when urea was close to depletion

225 (UA-T3). The *amt1* of *N. piranensis* was transcribed at high levels across all growth phases and
 226 conditions (Fig. 4a). It was most highly transcribed when growing on urea (46,483 TPM,
 227 Transcripts Per Million, UA-T1) and after transitioning to exponential growth on urea (50,880
 228 TPM, AU-T4). Transcript abundance dropped more than 7-fold within 24 hours after addition of
 229 ammonia to urea-grown cultures (6,498 TPM, UA-T2). The increased transcription of *amt1* in
 230 urea-grown *N. piranensis* (46,483 TPM, UA-T1) was consistent with our kinetics analyses
 231 showing a higher specific affinity for ammonia by urea-grown than ammonia-grown cells
 232 (28,066 TPM, AU-T1) (Fig. 3).

233 Unlike *N. piranensis*, no differential transcription ($P = 0.84$) of *amt1* (NVIE_002420) was
 234 observed between the exponentially growing *N. viennensis* on ammonia (1,379 TPM, AU-T1)
 235 and urea (1,348 TPM, UA-T1), consistent with the comparable affinity for ammonia using either
 236 ammonia- or urea-grown cells (Fig. 3). The transcripts of a *ut* (NVIE_014780) and *ureC* (urease
 237 alpha subunit, NVIE_014740) located within the same operon were ~10-fold higher or lower
 238 within 24 hours following addition of urea or ammonia, respectively (AU- & UA-T2 vs. T1).

239 Distinct from the AOA species, the comammox *N. inopinata* exhibited much less stringent
 240 control of urea consumption. Urea was hydrolyzed immediately upon its addition to ammonia-
 241 growing cells (Fig. 4c). As observed for the AOA species, *N. inopinata* significantly increased
 242 ($P < 0.01$) the transcription of its single *amt* when transitioning to exponential growth on urea
 243 upon ammonia depletion relative to growing on ammonia (AU-T3 vs. T1), while its transcript
 244 abundance was significantly lower ($P = 6.5 \times 10^{-27}$) 24 hours after ammonia addition to urea-
 245 grown cells (UA-T2 vs. T1). Its significantly higher transcription ($P = 2.8 \times 10^{-3}$) on urea (438
 246 TPM, UA-T1), versus on ammonia (312 TPM, AU-T1) is consistent with the higher ammonia
 247 affinity of cells precultured on urea (Fig. 3). Although the *urt* (urea transporter) and *ure*
 248 (NITINOP_3355–NITINOP_3362) were transcribed at significantly higher levels ($P = 7.0 \times 10^{-$
 249 ²⁵) during exponential growth on urea (UA-T3 vs. T1), the cells maintained relatively high
 250 transcription levels of these genes when grown on ammonia (Fig. 4c), which is distinct from

their extremely low transcription in the marine AOA *N. piranensis* before ammonia depletion (Fig. 4a). Comparable transcription of the *ure/ut* operon and an adjacent gene encoding a putative outer-membrane short-chain amides/urea porin (SAP, NITINOP_3364) in *N. inopinata* (Extended Data Fig. 2) suggested urea transport is regulated in both outer and inner membranes (Fig. 4c).

Since the substantial free ammonia released during consumption of urea by *N. lacus* would confound the effect of the added ammonia to urea-grown cultures, we instead examined transcript abundance at three time points of growth on an ammonia-urea mixture – exponential growth on urea (A+U-T1), near urea depletion (A+U-T2), and after shifting to exponential growth on ammonia (A+U-T3) (Fig. 4d). All genes involved in urea uptake and hydrolysis, including a putative *sap* (EBAPG3_007760), encoded in a single operon (EBAPG3_007760–EBAPG3_007800), were constitutively transcribed at relatively high levels at all time points, even when cells were using ammonia (Fig. 4d). This indicated that even while growing on ammonia, β -AOB *N. lacus* allocates substantial resources to facilitate use of its preferred N source urea.

Unlike the highly variable transcription of *amt* observed in the AOA and comammox, a single *N. lacus* Rhesus-type *amt* was always transcribed at low levels in *N. lacus* (Fig. 4d), as has also been observed in *Nitrosomonas* β -AOB and *Nitrosococcus* γ -AOB species^{26,27}. Of two genes encoding GlnB-type P_{II} proteins, one (EBAPG3_001140) was transcribed during all growth phases whereas a paralog (EBAPG3_010980) showed very limited transcriptional activity. In well characterized heterotrophs, the transport of ammonia by the AMT can be blocked by its binding to GlnB, controlled by uridylation/deuridylation post-translational modification of GlnB by GlnD in response to variations in intracellular energy and N status²⁸⁻³⁰. A *glnD* homolog in *N. lacus* (EBAPG3_003255) suggests this organism encodes a similar regulatory system. *GlnD* was transcribed at significantly higher levels ($P = 6.6 \times 10^{-15}$) when ammonia utilization was severely repressed (A+U-T1 and T2), suggesting that deuridylylated GlnB binds

to AMT and blocks ammonia transport in the presence of urea as shown by isotope tracking (Fig. 2).

Given the extremely slow growth of the marine γ -AOB *N. oceanii* on a urea-only medium (Fig. 1g), our transcript analysis only followed the response of cells growing on ammonia with subsequent urea addition. Urea hydrolysis started immediately upon addition and had no substantial effect on the growth rates or the transcription of genes directly involved in N species consumption (Fig. 4e), likely reflecting constitutive systems for ammonia and urea uptake without preference or adaptive lag.

Discussion

The AOM in our study set displayed a wide spectrum of adaptive strategies to coordinate growth on a mixture of urea and ammonia. Among characterized AOM species, AOA and comammox preferred ammonia, β -AOB preferred urea, and γ -AOB co-utilized ammonia and urea. The preferential use of ammonia is likely a common lifestyle shared among *Nitrosopumilus* and *Nitrososphaera* species, as the characterized genomes of these two AOA clades all encode the putative high-affinity AMT and GlnB-type P_{II} proteins suggested to be associated with efficient ammonia uptake and the repression of urea utilization in AOA, respectively (Extended Data Fig. 1a, Supplementary Discussion). Likewise, the genes involved in the repression of ammonia uptake in β -AOB (*glnB* and *glnD*) are universally present in *Nitrosospira* genomes (Extended Data Fig. 1b), suggesting a preference for urea may be common among ureolytic *Nitrosospira* species. In contrast, the *glnB* and *glnD* genes are largely absent in most *Nitrosomonas* genomes, which may in part explain the observed weaker repression of ammonia utilization by *Nitrosomonas ureae* compared to *Nitrosospira* species (Fig. 1).

Although distinct lineages of AOM often co-occur in the same environment and variable specific affinity for ammonia suggested to be the main determinant for competition and niche differentiation^{1,6-8,31-34}, our study indicates that differences in N substrate prioritization among

lineages may mitigate direct competition. For example, β -AOB with a high affinity for urea could prioritize their growth on urea (Fig. 3) while *Nitrosopumilus*/*Nitrososphaera* AOA would thrive in the ammonia-depleted environment due to their much higher affinity for ammonia. We can also now postulate that competitive advantage among *Nitrosopumilus* and related AOA is derived not solely from use of both N substrates, but also from the adaptive capacity of marine AOA to increase their affinity for ammonia upon ammonia exhaustion (Fig. 3), resulting in more efficient scavenging of ammonia released from organic matter decomposition by other microorganisms^{35,36}.

One of the most significant findings of our study is that ammonia oxidation and ammonia assimilation are both transport-dependent. Our stable isotope tracking analysis clearly demonstrated that β -AOB did not oxidize extracellular ammonia when growing on a mixture of ammonia and urea. Rather, only ammonia originating in the cytoplasm from urea hydrolysis was oxidized. Ammonia transport is blocked when N needs are satisfied by cytoplasmic hydrolysis of urea, most likely by binding of a GlnB-type P_{II} regulatory protein to the AMT. Thus, these data also implicate an undescribed mechanism for direct or facilitated delivery of cytoplasmic ammonia to the AMO active site.

Since extracellular ammonia must be transported into the cytoplasm by the AMT before being oxidized by the AMO, the AMT must be the rate-limiting functional unit that determines affinity for both the assimilation and oxidation of ammonia among AOM. Although recently determined K_m values of ammonia transporters (e. g., ~800 μ M for *E. coli* AMT, 3.9 mM at pH 7 for *Archaeoglobus fulgidus* Amt1³⁷⁻³⁹) are higher than the cellular $K_{m(app)}$ determined in our study, kinetic data for ammonia assimilation by environmental populations of phytoplankton or heterotrophic bacteria indicates the existence of ammonia transporter variants with higher affinities ($K_{m(app)}$ 0.1–10 μ M)^{40,41}. Combined modelling and substrate uptake studies of oligotrophic marine bacteria have demonstrated a relationship between the number of cell surface transporters and a° ^{42,43}. A recent extensive investigation of apparent affinity of different

AOM lineages showed that the affinities for ammonia correlate with their cell surface area to volume ratios⁸, suggesting that a higher cellular affinity for ammonia is achieved by the higher number of AMTs that can be accommodated on a larger surface area. In our study, a 5.1-fold increase in the a° of *N. piranensis* for ammonia was correlated with increased transcription of its high-affinity AMT (Figs. 3b and 4a). In contrast, the transcription of *amo* remained relatively unchanged during its exponential growth on either N substrate (AU-T1 vs. UA-T1, Supplementary Table 3). Since it is unlikely that a single species can substantially change its cell surface area to volume ratio, the increased cellular ammonia affinity observed for urea-grown *N. piranensis* and *N. inopinata* is most likely caused by the increase in the density of high-affinity AMT per cell surface area.

We recognize that AOM pure cultures representing a broader range of ecotypes are needed to further explore the diversity of N substrate preference and supporting regulatory systems among globally abundant ammonia oxidizers. For example, unlike other AOA lineages, *Nitrosocosmicus* species lack the putative high-affinity AMT (Extended Data Fig. 1a) and exhibit highly active urea hydrolysis activity when grown in urea-only media⁴⁴, which suggests their regulation of N substrate utilization is distinct from *Nitrosopumilus* and *Nitrososphaera* species. In addition, while both clade A and clade B comammox species possess ATP-dependent urea transporters, they differ in their encoded ammonia transporter types and the presence of GlnD proteins⁴⁵ (Extended Data Fig. 1c). As more diverse comammox strains become available, it will be interesting to see whether different transporter inventories impact the regulatory mechanisms controlling N catabolite repression.

In summary, our study revealed an unexpected diversity in substrate preference and supporting regulatory systems among AOM and has significantly changed our understanding of their ecophysiology and niche differentiation. Our surprising finding that ammonia oxidation is transport-dependent suggests the need for modifying current models of ammonia oxidation. Our finding that cellular ammonia affinity is not an intrinsic property of AOA and comammox

354 species, but a property that can be adaptively regulated in response to the environmental
355 ammonia availability, has implications on how N resource competition is modelled among AOM
356 groups. Adding these novelties to models will ultimately offer a more holistic and accurate
357 representation of N resource partitioning among AOM in natural and engineered systems.

Methods

Laboratory strains: *Nitrososphaera viennensis* strain EN76 was obtained from the German Type and Culture Collection (DSMZ, Braunschweig, Germany). *Nitrosopumilus piranensis* strain D3C was kindly provided by Dr. Alyson Santoro and Dr. Barbara Bayer (University of California, Santa Barbara), and *Nitrosomonas ureae* strain Nm 10 was kindly provided by Dr. Andreas Pommerening-Röser (University of Hamburg). *Nitrospira multiformis* strain ATCC25196 was kindly provided to University of Oklahoma by Dr. Jeanette Norton (Utah State University). *Nitrospira lacus* strain APG3 was isolated from Green Lake in Seattle, Washington. *Nitrosococcus oceani* strain ATCC 19707 was obtained from the American Type Culture Collection (ATCC). *Nitrospira inopinata* strain ENR4 was kindly provided to University of Washington by Dr. Holger Daims (University of Vienna). For measurements (such as the affinity measurements) at all other locations, *N. inopinata* was obtained from DSMZ.

Cell maintenance: Pure cultures of marine AOA *N. piranensis* and marine γ -AOB *N. oceani* were maintained in HEPES-buffered Synthetic Crenarchaeota Medium (SCM), and soil AOA *N. viennensis*, comammox *N. inopinata*, β -AOB *Nitrososphaera lacus*, *Nitrospira multiformis*, and *Nitrosomonas ureae* were maintained in HEPES-buffered Freshwater Medium (FWM)^{46,47}. All media pH were 7.6–7.8. All stock cultures were grown in 250 mL bottles containing 100 mL medium supplemented with ammonia or urea in the dark to avoid light inhibition. *N. viennensis* and *N. piranensis* were additionally supplemented with 100 μ M sodium pyruvate as a scavenger for hydrogen peroxide (H₂O₂) or other reactive oxygen species. All cultures were grown without agitation except *N. oceani*, which was grown with a shaking speed of 70 r.p.m. *N. piranensis*, *N. viennensis*, *N. inopinata*, *N. lacus*, *N. multiformis*, *N. ureae*, and *N. oceani* were incubated at 30 °C, 42 °C, 37 °C, 20 °C, 28°C, 25 °C, and 25 °C, respectively. The medium was pre-warmed to culturing temperature before inoculation to avoid lower growth rates induced by temperature shock. Growth was monitored by ammonia or urea consumption as well as nitrite or nitrate

accumulation. When mid- to late-exponential phase was reached, 0.5–1% inoculum was transferred to fresh media.

Culture purity was confirmed monthly by absence of growth on R2A agar and (marine) LB broth. Sanger sequencing of 16S rRNA genes was conducted every 6 months to confirm purity of the strains.

Growth experiments on single and mixed N substrates

Growth experiments: Incubation experiments were set up with three different conditions: ammonia-only, urea-only, and ammonia plus urea. Inoculation was done with 0.5–1% transfer from the maintenance culture at mid-exponential phase. The inoculum used for the ammonia-only and urea-only experiments were transferred from maintenance culture growing in media containing ammonia or urea as the only N substrate, respectively. The ammonia plus urea experiment was conducted with both ammonia- and urea-grown maintenance cultures. Media composition, culture volume, and growth conditions were the same as for the maintenance cultures. A negative control bottle was prepared for the urea-only and ammonia plus urea treatments without addition of cells to confirm negligible non-biological degradation of urea during experimental periods. Samples were taken roughly at every doubling time for nutrient measurements (urea, NH_3 , NO_2^- , and NO_3^-). Specific growth rates (μ) were determined from the log linear region of NO_2^- or $\text{NO}_2^- + \text{NO}_3^-$ production for AOA/AOB or comammox, respectively (Supplementary Fig. 5). AOM (ammonia-oxidizing microorganisms) culture purity checks were performed at the end of each growth experiment. No growth was detected on R2A agar or (marine) LB broth, indicating the absence of contaminants. In addition, transcriptomics data analysis did not identify any contaminant transcripts.

Cell count: A mounting medium was prepared consisting of 3 μL of 1 mol/L ascorbic acid and 4 μL of SybrGreen I stock solution (10,000-fold concentrated in DMSO, equivalent to 10 mmol/L) in 200 μL of Moviol solution (Moviol 4-88; Fluka)⁴⁸. Cells were captured on a 0.02- μm

408 Anopore membrane filter (Whatman) and stained with 20 μ L of the mounting medium. Using a
409 Zeiss Axioskop 2 MOT epifluorescence microscope (Carl Zeiss AG, Germany) (for freshwater
410 and soil AOM experiments) and Nikon Eclipse Ni microscope (Nikon Instruments Inc., Japan)
411 (for marine AOM experiments), cell numbers were determined for 20 random fields (20–100
412 cells per field) per sample.

413 No discernible large cell aggregates were observed for *N. piranensis*, *N. viennensis*, *N. inopinata*,
414 *N. lacus*, and *N. ureae*, under the tested growth conditions in this study. Cell aggregates were
415 occasionally observed for *N. oceanii* growing in urea-only media, which could be attributed to
416 extremely slow growth on urea (Fig. 1g) and potential stress response. To ensure accurate cell
417 counts, the urea-grown *N. oceanii* cultures were meticulously pipetted up and down to disperse
418 any possible cell aggregates before harvesting cells for cell counts. Some cell aggregates were
419 observed in growth experiments of *N. multiformis* without agitation, which may be attributed to
420 oxygen limitation caused by rapid O₂ consumption during high rates of ammonia oxidation. For
421 cell counts, each culture sample was passed through a syringe and needle (0.4 mm diameter) at
422 least three times to disperse aggregated cells, following the previously described method ⁴⁴. In
423 contrast, no aggregates were observed during kinetics measurements (see details below) using
424 microrespirometry chambers with continuous stirring to maintain O₂ homogeneity.

425 Extracellular urease activity check: Incubation experiments were conducted to check whether
426 urease activities occurred extracellularly. Cultures were grown to early stationary phase in urea-
427 only media using the same volume and growth conditions as the maintenance cultures. The same
428 cultures were then filtered through 0.2 μ m PES filter membrane (Pall Supor[®]), preventing AOM
429 cells from passing through and leaving only remaining active urease enzymes, if any, in the
430 filtrate. The filtrates were supplemented with additional urea, and concentrations of ammonia
431 and urea were monitored for 1–2 weeks to check for extracellular urease activity.

Chemical analysis: Nitrite was measured spectrophotometrically using the sulphanilamide and N-(1-naphthyl)-ethylenediamine dihydrochloride method (EPA Method 354.1). Ammonia concentrations were either determined colorimetrically from the monochloramine generated by the reaction with alkaline hydrolysis of sodium dichloroisocyanurate (ISO 15923-1) or using the o-phthaldialdehyde fluorescence method ⁴⁹. For determination of nitrate, NO₃⁻ was first reduced to NO₂⁻ by hydrazine or vanadium chloride ⁵⁰, then the procedure for NO₂⁻ measurement was followed to determine the total oxidized N (TON). NO₃⁻ was determined from TON minus NO₂⁻. Urea was measured according to the previously described protocols ⁵¹. First, an aliquot was split from the original sample and treated with urease (Millipore Sigma #U1500) to convert urea into ammonia. Ammonia was measured for the urease-treated aliquot and the original sample. Urea-N was then determined by subtracting the ammonia concentration in the original sample (untreated) from the aliquot (urease-treated).

¹⁵N label tracking for ammonia oxidation

Incubation with labelled substrates: Cells were incubated as described in “Cell maintenance” with additional details provided in this section. All incubations were done in ammonia plus urea medium (1:1 N molar ratio) but with different substrate labeled. One set of experiments was conducted with ~5% ¹⁵N-ammonium chloride (Sigma-Aldrich #299251) (and 95% unlabeled ammonia) for the *N. piranensis*, *N. viennensis*, *N. lacus*, and *N. oceani* incubations. The other set of experiments was conducted with ~5% ¹⁵N-urea (Sigma-Aldrich #316830) (and 95% unlabeled urea) for these four AOM species. Given that no observable effect on the ammonia oxidation activity was detected when varying ¹⁵N labeling percentages from 1 to 10% (Supplementary Fig 6), the labelling incubation experiments for *N. inopinata* were conducted using a lower ¹⁵N labeling percentage (~2.5% ¹⁵NH₄Cl or ~2.5% ¹⁵N-urea) compared to the other tested AOM species, which allowed us to obtain a more appropriate range of δ¹⁵N signals for accurate measurement using the GC-IRMS system (see detailed methods below). Three or four parallel incubations were set up for biological replicates and nutrient measurements. Four milliliters of

458 cultures were collected at each time point. Two milliliters of culture were used for nutrient
 459 measurements, and the other 2 mL of cultures were filter-sterilized using a 0.2 μm syringe-
 460 driven filter into a microcentrifuge tube and immediately frozen at $-80\text{ }^{\circ}\text{C}$ for isotope analysis.

461 $\delta^{15}\text{N}$ of NH_4^+ , NO_2^- , and NO_3^- : $\delta^{15}\text{N}$ of NO_2^- for *N. lacus* and *N. oceani* incubations was
 462 measured by chemical conversion (sodium azide, Sigma-Aldrich) of NO_2^- to N_2O ⁵². $\delta^{15}\text{N}$ of
 463 NO_x^- (NO_2^- plus NO_3^-) for *N. inopinata* incubation was determined using the bacterial denitrifier
 464 method ⁵³. $\delta^{15}\text{N}$ of the converted N_2O either through chemical conversion or denitrifier
 465 reduction was quantified using a GC-IRMS system (a cryogenic extraction and purification
 466 system interfaced to a Delta V isotopic ratio mass spectrometer) ⁵⁴. $\delta^{15}\text{N}$ of NO_2^- was calibrated
 467 against five in-house NO_2^- standards ($\delta^{15}\text{N}$ of the five in-house NO_2^- standards were 5.19‰,
 468 126.07‰, 336.54‰, 671.78‰, 1343.28‰, respectively). $\delta^{15}\text{N}$ of NO_3^- values were calibrated
 469 against NO_3^- isotope standards USGS 34, IAEA N3 and USGS 32. Standard curves were run at
 470 the beginning and end of sample analysis and at ten sample intervals.

471 For $\delta^{15}\text{N}$ of NH_4^+ , the samples were measured using hypobromite oxidation coupled to azide
 472 conversion method ⁵⁵ with slight modification ⁵⁶. Before the NH_4^+ was oxidized to NO_2^- , the
 473 pre-existing NO_2^- was removed from samples using sulfamic acid ($\geq 99\%$ sulfamic acid, Sigma-
 474 Aldrich) ⁵⁷. The removal efficiency of NO_2^- was tested by measuring the NO_2^- concentration
 475 before and after adding the sulfamic acid to $10\text{ }\mu\text{mol L}^{-1}$ of NO_2^- , and the oxidation efficiency of
 476 NH_4^+ to NO_2^- was tested by comparing the NH_4^+ and NO_2^- concentrations before and after
 477 hypobromite oxidation with initial NH_4^+ concentration of $2\text{ }\mu\text{mol L}^{-1}$. The NO_2^- removal and
 478 NH_4^+ oxidation efficiencies in our experiments were $>99\%$ and $>90\%$, respectively. All the
 479 samples were diluted to the NH_4^+ concentration of $2\text{ }\mu\text{mol L}^{-1}$ for hypobromite oxidation. $\delta^{15}\text{N}$
 480 of NH_4^+ was calibrated against five in-house NH_4^+ standards ($\delta^{15}\text{N}$ of the five in-house NH_4^+
 481 standards were 0.31‰, 135.04‰, 326.07‰, 588.08‰, 1324.84‰, respectively). Standard
 482 curves were run at the beginning and end of sample analysis and at ten sample intervals.

Contribution of NO_x^- from NH_4^+ and urea: The contribution of NH_4^+ (or urea) to NO_2^- production for *N. lacus* and *N. oceani* incubations, and to NO_x^- production for *N. inopinata* incubation was calculated based on the isotopic mass balance model at each time point. Briefly, $\delta^{15}\text{N}$ of the net produced NO_2^- between two consecutive time points during the time course incubation was calculated using a two-end member mixing model (Equation 1).

$$\delta^{15}\text{N}_{\text{NO}_2} = \frac{C_{t2} \times \delta^{15}\text{N}_{\text{NO}_2 \text{ } t2} - C_{t1} \times \delta^{15}\text{N}_{\text{NO}_2 \text{ } t1}}{C_{t2} - C_{t1}} \text{ (Equation 1)}$$

where $\delta^{15}\text{N}_{\text{NO}_2}$, $\delta^{15}\text{N}_{\text{NO}_2 \text{ } t2}$, and $\delta^{15}\text{N}_{\text{NO}_2 \text{ } t1}$ denote the $\delta^{15}\text{N}_{\text{NO}_2^-}$ value of the net produced NO_2^- during the $^{15}\text{NH}_4^+$ or ^{15}N -urea labelling incubation, $\delta^{15}\text{N}_{\text{NO}_2^-}$ at two consecutive time points, respectively. C_{t2} and C_{t1} denote the measured NO_2^- concentration at the two time points, respectively.

The fraction contribution of the net produced NO_2^- by NH_4^+ and urea between two consecutive time points were derived using Equation 2.

$$\text{AT}^{15}\text{N}_{\text{NO}_2^-} = \text{AT}^{15}\text{N}_{\text{NH}_4^+} * a\% + \text{AT}^{15}\text{N}_{\text{urea}} * (100\% - a\%) \text{ (Equation 2)}$$

where $\text{AT}^{15}\text{N}_{\text{NO}_2^-}$ is the fraction of ^{15}N atom (%) of the net produced NO_2^- between two time points, which is calculated by the $\delta^{15}\text{N}_{\text{NO}_2}$ in the Equation 1. $\text{AT}^{15}\text{N}_{\text{NH}_4^+}$ and $\text{AT}^{15}\text{N}_{\text{urea}}$ is the fraction of ^{15}N atom (%) in the NH_4^+ and urea pool of the corresponding time point (refers to $t1$ in Equation 1). $a\%$ denotes the fractional contribution of NH_4^+ to the net produced NO_2^- at the corresponding time point.

Tracing the sources of N and C for assimilation by NanoSIMS analysis

Incubation with labeled substrates: Cells were incubated as described in “Cell maintenance” except as detailed in this section. All incubations were done in ammonia plus urea medium (1:1 N molar ratio) but with different substrate labeled. One set of experiments used ^{15}N -ammonium chloride (Sigma-Aldrich #299251) and ^{13}C -sodium bicarbonate (Cambridge Isotope Laboratories

506 #CLM-441-5), and the other set of experiments used $^{15}\text{N}^{13}\text{C}$ -dual labeled urea (Sigma-Aldrich
507 #490954). We used 100% labeled compounds for *N. viennensis*, *N. inopinata*, *N. lacus*, and *N.*
508 *oceani* incubations. Since the fully labeled compounds inhibited the growth of *N. piranensis*,
509 30% labeled compounds was used.

510 For *N. viennensis*, *N. inopinata*, and *N. lacus*, cells were harvested for the first time point when
511 the preferred N substrate was consumed and for the second time point when both N substrates
512 were depleted. For *N. piranensis*, cells were collected for the first time point during the
513 exponential growth using ammonia and the second time point when both ammonia and urea were
514 exhausted. For *N. oceani*, which co-utilized ammonia and urea, cells were harvested at late
515 exponential phase and stationary phase for the first and second time points, respectively. Growth
516 was monitored by measuring ammonia and urea consumption and nitrite/nitrate accumulation.
517 Eight to twelve parallel incubations were set up for biological replicates, and cells were collected
518 and fixed ($> 9 \times 10^6$ cells) from 3–4 replicate cultures at each time point for NanoSIMS analysis.
519 An unlabeled control was prepared with all unlabeled media components, and a killed control
520 was prepared with all labeled media components and a 2% (v/v) paraformaldehyde-killed
521 inoculum. Before adding the killed inoculum, cells were pelleted at $10,000\times g$ at 20°C for one
522 hour and re-suspended in a standard 1X phosphate buffered saline (PBS) solution. This step was
523 repeated two more times to remove residual paraformaldehyde.

524 Cells were collected and fixed in 2% (v/v) paraformaldehyde at room temperature in the dark for
525 30–40 mins, diluted with sterilized Milli-Q water for a target of 10 million cells, then filtered
526 onto a 25 mm diameter $0.2\ \mu\text{m}$ pore size polycarbonate membrane (Millipore Sigma
527 #GTTTP02500) using vacuum filtration. Cells were then washed three times with sterile filtered
528 Milli-Q water to remove residual labeled media. The filter was put in a single well of a 12 well
529 plate (sample side facing up) (Thermo Fisher Scientific, Rochester, NY, USA) and stored at 4°C
530 in the dark for following NanoSIMS analysis.

531 DIC measurement: Three media samples for both labeled treatments (30 mL), an unlabeled
532 media control sample, and an atmospheric gas control sample were collected in 50 mL sealed
533 serum bottles for DIC analysis. In addition, 3 unfixed culture samples were taken at each time
534 point and stored at 4°C for ^{13}C DIC measurement. These liquid samples were acidified with HCl,
535 driving all DIC into CO_2 . Headspace gas was collected for isotopic gas analysis using a Picarro
536 G2201-I Cavity Ring Down Spectrometer (Picarro Inc., Santa Clara, CA, USA). Vials were
537 initially autoclaved and filled with one atmosphere of nitrogen gas prior to 5mL of gaseous
538 sample headspace addition. Each vial then had 5mL of gaseous headspace removed using a
539 gastight syringe injected into the Small Sample Introduction Module 2. The factory protocol was
540 then run with a zero-air dilution and analyzed for $^{13}\text{CO}_2/^{12}\text{CO}_2$. Ambient air was used between
541 every third injection as a reference point for calibration and to ensure lack of instrument drift.

542 NanoSIMS: Filters were cut into wedges of 1/8 size using sterile scissors and adhered to
543 aluminum disks using conductive tabs (#16084-6, Ted Pella, Redding, CA) and sputter coated
544 with ~5 nm of gold. Isotope imaging was performed with a CAMECA NanoSIMS 50 at
545 Lawrence Livermore National Laboratory. The primary $^{133}\text{Cs}^+$ ion beam was set to 1.5 pA,
546 corresponding to an approximately 150 nm diameter beam diameter at 16 keV. Rastering was
547 performed over $20 \times 20 \mu\text{m}$ analysis areas with a dwell time of 1 ms pixel^{-1} for 19-30 scans
548 (cycles) and generated images containing 256×256 pixels. Sputtering equilibrium at each
549 analysis area was achieved with an initial beam current of 90 pA to a depth of ~60 nm. After
550 tuning the secondary ion mass spectrometer for mass resolving power of ~7000 (1.5 $\times$ corrected),
551 secondary electron images and quantitative secondary ion images were simultaneously collected
552 for $^{12}\text{C}_2^-$, $^{13}\text{C}^{12}\text{C}^-$, $^{12}\text{C}^{14}\text{N}^-$, and $^{12}\text{C}^{15}\text{N}^-$ on individual electron multipliers in pulse counting mode;
553 $^{13}\text{C}^{12}\text{C}^-/^{12}\text{C}_2^- = 2 \cdot ^{13}\text{C}/^{12}\text{C}$ and $^{12}\text{C}^{15}\text{N}^-/^{12}\text{C}^{14}\text{N}^- = ^{15}\text{N}/^{14}\text{N}$ ⁵⁸. All NanoSIMS datasets were
554 initially processed using L'Image (<http://limagesoftware.net>) to perform dead time and image
555 shift correction of ion image data before creating $^{13}\text{C}/^{12}\text{C}$ and $^{15}\text{N}/^{14}\text{N}$ ratio images, which
556 reflected the level of ^{13}C and ^{15}N incorporation into biomass. To quantify substrate

incorporation, cells were identified based on $^{12}\text{C}^{14}\text{N}^-$ images, and regions of interest (ROIs) for cells were drawn with the automated algorithm. The automatic particle analysis algorithm has no difficulty in separating cells from one another. The isotopic ratio averages and standard error of the mean for each ROI were calculated from ratios by cycle. Excess fraction of cellular C and N incorporated in cells from the enriched isotopic substrate (C_{net} and N_{net} , respectively) was calculated based on the isotope fraction in a target cell (F_c), the killed controls (F_{kc}) to represent the initial enrichment of the cells, and the isotope fraction of the added substrates (F_s), according to the equation (Equation 3) ⁵⁹:

$$[C \text{ or } N]_{\text{net}} = \frac{F_c - F_{kc}}{F_s - F_{kc}} \text{ (Equation 3)}$$

The ROIs data were exported for further statistical analyses in R, which was used to calculate the mean and standard error of the mean C_{net} and N_{net} for each treatment and timepoint. Subsequently, % new N incorporation from urea or ammonia, and % new C incorporation from urea or bicarbonate were calculated, e.g. % new N incorporation from urea = $N_{\text{neturea}} / (N_{\text{neturea}} + N_{\text{netammonia}})$.

Affinity and kinetics measurements

Cell maintenance: Cells were maintained as described in “Cell maintenance” except transfer was done with 5% v/v inoculum and culture volume was 200 mL in 1 L glass bottles. Each strain was maintained with either 1.0 mM NH_4Cl or 0.5 mM urea and were used separately to determine oxidation kinetics. *N. piranensis* and *N. viennensis* cultures were additionally supplemented with 100 μM α -ketoglutaric acid and 500 μM sodium pyruvate, respectively, as H_2O_2 scavengers.

Substrate-dependent oxygen uptake: Substrate oxidation kinetics were measured using a microrespirometry (MR) system setup described previously ⁶⁰. Briefly, the setup was equipped with 80 mL three-port glass MR chambers (Unisense AS, Denmark), a magnetic stir bar, and an

581 OX-MR oxygen microsensor (Unisense AS, Denmark). Sensors were polarized and pA signals
582 were amplified and converted to mV using PA2000 picoammeters (Unisense AS, Denmark).
583 Data were recorded at 1 Hz using Lab-Trax-4/16 data acquisition system (World Precision
584 Instruments, Sarasota, FL) and LabScribe version 4.0 (iWorks, Dover, NH). Oxygen sensors
585 were continuously polarized before use and calibrated using culture media equilibrated with
586 laboratory air and oxygen-free water in a 4 mL calibration chamber (Unisense AS, Denmark) for
587 zero reading according to manufacturer's recommendations. All measurements were done in a
588 re-circulated water bath at the optimal growth temperature for each strain.

589 Nitrite concentrations in cell cultures were monitored daily, and early-stationary phase cells
590 (culture pH 7.3–7.5) were transferred directly into the MR chamber upon N substrate depletion
591 (of ammonia or urea) to minimize perturbations as previously described ⁶¹. *N. inopinata* cells
592 were transferred when both NH₃ and NO₂⁻ were depleted. The chamber was filled with culture
593 to overflowing and closed with MR injection lids to ensure that no headspace was left. For *N.*
594 *viennensis*, 50 mg L⁻¹ kanamycin was added to the chamber after culture transfer to avoid
595 bacterial contamination. Background sensor signal drift was measured for 30 to 60 mins prior to
596 initial substrate injections. Cells were initially activated by addition of 5 μM NH₄Cl or 5 μM
597 urea, depending on which substrate cells were grown with. To obtain O₂ uptake traces for
598 kinetic parameter determination, 20–40 μM NH₄Cl were injected next, and O₂ uptake was
599 recorded until at least 30 mins after substrate depletion. Subsequently, 10–20 μM urea was
600 injected for urea-dependent kinetics measurements. For *N. lacus* and *N. multiformis*,
601 measurements described above yielded no discernable activities. Therefore, measurements with
602 discrete substrate additions were conducted as follows. Initial O₂ uptake was recorded for at
603 least 30 mins with multiple ammonia (5 μM–10 mM) or urea (2.5–500 μM) injections to cover
604 the large substrate concentration range required for determination of kinetic constants in these
605 strains.

Cell counting and estimation of protein content: Cells used for MR measurements were filtered onto 0.2 µm black polycarbonate membranes (MilliporeSigma, Germany) for cell counts. Filters were air-dried and mounted with SYBR Green I (Invitrogen, Waltham, MA) in Moviol mounting solution, and incubated at -20 °C for at least 2 hours and counted by epi-fluorescence microscopy⁴⁸. Cell volumes were calculated by previously reported cell diameter and length of each strain^{4,15,20,47,62}. Protein contents for each strain were estimated based on protein to cell volume ratios⁶³.

Calculation of kinetic constants: O₂ uptake traces were used to determine kinetic constants for the tested AOM strains. Substrate to O₂ consumption stoichiometry ratios of 1:1.5 using total ammonia (NH₃ + NH₄⁺), and 1:3 using urea were used to calculate N substrate concentrations at given O₂ uptake rate. For *N. inopinata*, the expected stoichiometry of total ammonia to O₂ was 1:2⁷. However, in this study we found that at low ammonia concentrations, *N. inopinata* oxidized NH₃ only to NO₂⁻, and not NO₃⁻, and hence only 1.5 mol of O₂ was consumed per mol N of NH₃ or urea substrate under measurement conditions, where NO₂⁻ concentrations remained far below its $K_{m(app)}$ for NO₂⁻. Kinetic constants were obtained by fitting a Michaelis-Menten equation to total ammonia or urea and oxidation rates using the following equation (Equation 4):

$$V = (V_{max} \times [S]) \times (K_{m(app)} + [S])^{-1} \text{ (Equation 4)}$$

where V refers to the rate of oxygen uptake in µmol mg protein⁻¹ h⁻¹, V_{max} is the maximum rate of oxygen uptake (µmol mg protein⁻¹ h⁻¹), S is the concentration of substrate (µM), and $K_{m(app)}$ is the concentration of N substrate at which half of the maximum rate is achieved (µM). $K_{m(app)}$ and V_{max} values for total ammonia and urea were calculated in NH₃ and urea-grown strains, respectively. Specific affinities were calculated from kinetic constants of each strains using the equation (Equation 5):

$$a^0 = V_{max} \times K_{m(app)}^{-1} \text{ (Equation 5)}$$

Here, a^0 is specific affinity ($\text{L g wet cell}^{-1} \text{h}^{-1}$), V_{max} values were normalized to protein concentration ($\mu\text{mol N mg protein}^{-1} \text{h}^{-1}$), and $K_{\text{m(app)}}$ is the apparent half-saturation constant ($\mu\text{mol L}^{-1}$). The factor of 5.7 g wet weight per g protein was used to estimate wet weight from protein content^{61,64}. All measurements were carried out in at least ten replicates to obtain at least three replicates with low enough noise for subsequent calculation of kinetic constants. Each trace was manually inspected for stirring noise and only the 3 to 4 traces with least noise were subjected to downstream kinetic analyses. Replicates for each strain were generated over a two-to-four-week timeframe with no more than two out of ten traces originating from the same culture. Kinetic constants ($K_{\text{m(app)}}$ and a^0) ranged over several orders of magnitude and Levene's test showed that residuals were not homogenous. Therefore, $K_{\text{m(app)}}$ and a^0 data were log₁₀-transformed to satisfy parametric test assumptions. Comparisons of means were then conducted by one-way analysis of variance (ANOVA). Kinetic constants for ammonia in ammonia-grown strains and kinetic constants for ammonia and urea in urea-grown strains were compared as a factor, yielding 12 groups. Statistically different groups were determined by Tukey's HSD post-hoc analysis ($\alpha = 0.05$) with Bonferroni correction. Statistical analyses were performed using SPSS 21 (IBM SPSS Inc., 2016) and R version 3.6.3.

Transcriptomics analysis

Alternative nitrogen substrate spiking experiment: To capture the regulatory response of AOM to switches between ammonia and urea utilization, incubations were conducted with one initial N substrate and spiked with the second N substrate via two directions: (1) incubation with initially ammonia then spiked with urea (AU) and (2) incubation with initially urea then spiked with ammonia (UA). Because ammonia was secreted out of the cells during urea consumption in *N. lacus* incubations, *N. lacus* was instead grown on a mixture of ammonia and urea (A+U). All media and growth conditions were the same as described in the section "Cell maintenance" except incubations were done in larger volumes to obtain enough cell material for RNA extraction. For each experiment, 12–18 parallel incubations of 400 mL cultures were grown in 2

656 L bottles. One exception was *N. lacus*, which was grown in 800 mL culture volumes in 2 L
657 bottles to maximize cell material. Preliminary incubation was conducted with 800 mL *N. lacus*
658 cultures to confirm no oxygen limitation. The inoculum used for the AU experiments were
659 transferred from ammonia-grown maintenance culture, whereas urea-grown maintenance culture
660 was used as the inoculum for the UA experiments.

661 Cell harvesting: For each replicate of RNA sample, the entire culture from 1–2 bottles (each with
662 400 mL culture) was filtered onto a Supor® 0.2 µm 47mm membrane filters (Pall Corporation,
663 N.Y.). The filters were immediately flash frozen and stored at -80 °C until RNA extraction.
664 Each time point had 4 replicate RNA samples. Due to the differences in physiological response
665 to alternative N substrate addition, the collected time points differed for each AOM and are
666 summarized in Supplementary Table 4.

667 RNA extraction and RNA-sequencing: For each time point, RNA was extracted from four
668 biological replicates following a manual phenol-chloroform RNA extraction method as
669 previously described⁶⁵. Briefly, the entire membrane filter (with cells collected) was dissolved
670 in an extraction buffer containing 21 mM sodium acetate, 0.6 mM EDTA, 0.6% SDS, and 51%
671 (v/v) phenol solution preheated to 80 °C (saturated with 0.1M citrate buffer, pH 4.3, Fisher
672 Scientific, USA). The samples were disrupted with a bead beater (MP Biomedical) at 4 m/s for
673 30 seconds and spun down. The supernatant was treated with equal volume of
674 phenol/chloroform/isoamyl alcohol (125:24:1) (pH 4.5, Fisher Scientific, USA). The bead
675 beating, spin down, and treatment steps were repeated with chloroform:Isoamyl alcohol (24:1)
676 (Sigma Aldrich, USA). The final supernatant was treated with 0.8M of lithium chloride and
677 equal volume of 100% isopropanol and precipitated for RNA at -80 °C overnight. The
678 precipitant was pelleted with centrifugation, washed with 75% ethanol, and dried at room
679 temperature. The RNA pellet was then dissolved in RNase free water. The RNA sample was
680 treated twice with Ambion's DNA-freeTM DNA removal kit, and the precipitation steps with
681 lithium chloride and overnight precipitation at -80 °C were performed again. The final RNA was

recovered using the MP spin column (MP Biomedical, USA), washed twice with 75% ethanol, and eluted with RNase free water.

The DNA contaminant removal efficiency was verified by qPCR analysis showing no or very low amplification copy number. The concentrations and quality of RNA samples were checked with an Agilent 2100 bioanalyzer. The three replicates with the highest RNA integrity numbers were shipped to Azenta Life Science (New Jersey, USA) for rRNA depletion with the QiaSeq FastSelect kit (Qiagen, USA, Cat. No.: 335921), library preparation, and sequencing on the Illumina HiSeq 2000 platform. About 18 RNA samples were sequenced per HiSeq lane at roughly 5GB data per sample.

Gene transcription analysis: Raw reads were trimmed by Trimmomatic (version 0.36)⁶⁶, and clean reads were aligned with the reference AOM species genomes using Bowtie2 (version 2.3.4)⁶⁷. The alignments were postprocessed into sorted BAM files with Samtools (version 0.1.19)⁶⁸. Reads were attributed to open reading frames (ORFs) using the htseq count tool from the HTseq 2.0⁶⁹. TPM (Transcripts per million) values were calculated as follows (Equation 6):

$$TPM = \frac{rg \times rl \times 10^6}{cl \times T} \text{ (Equation 6)}$$

where rg is reads mapped to gene g , rl is read length, cl is coding sequence (CDS) length, and T is sum of $rg \cdot rl \cdot cl^{-1}$ for all genes. Differential abundance analysis was performed with DESeq2 (version 1.2.10)⁷⁰.

Comparative genomics analysis

All publicly available and published AOM genomes were downloaded from the NCBI Refseq database, as of June 7th, 2023. Only high-quality genomes with a completeness $\geq 90\%$ and contamination $\leq 5\%$ ⁷¹ were chosen, and the selected genomes were further clustered using a multistep distance-based approach with an average nucleotide identity (ANI) threshold of 95% over at least a 30% alignment fraction by dRep (version 3.4.2)⁷². From each cluster, the best

quality genome was selected as the representative for comparative analysis, considering factors such as genome completeness, minimal contamination, and assembly N50 (with culture genomes always given preference over metagenome-assembled genomes). The encoded protein sequences of 136 AOA, 92 AOB, and 61 comammox culture genomes and metagenome-assembled genomes were searched via diamond (version 0.9.24.125) ⁷³ with an E-value of $\leq 1 \times 10^{-5}$, identity of $\geq 50\%$, and coverage of $\geq 50\%$ against an in-house database of ammonia and urea uptake and utilization genes found in KEGG database. Completeness and contamination levels of AOM genomes were assessed by CheckM (version 1.0.18) ⁷⁴. SignalP 6.0 ⁷⁵ was utilized to predict the presence of signal peptides in the urease enzyme of the tested AOM species. The phylogenomic trees of AOA, AOB, and comammox species were constructed based on concatenated alignments of marker proteins from GTDB database ⁷⁶. The concatenated alignments were constructed by GTDB-Tk (version 2.0.0) ⁷⁷, and the trees were built using IQ-TREE (v. 2.1.2 COVID-edition) ⁷⁸ with 1,000 times ultrafast bootstrapping.

Statistics & Reproducibility

Statistics for kinetic analysis were performed using one-way analysis of variance (ANOVA), and differential abundance analysis for transcriptomics studies was performed with DESeq2 (version 1.2.10) ⁷⁰. No statistical method was used to predetermine sample size. No data were excluded from the analyses. The experiments were not randomized. The investigators were not blinded to allocation during experiments and outcome assessment.

Data Availability: All data are available in the main text or the supplementary materials. RNA-seq data in this study have been deposited in the NCBI GEO database under BioProject ID PRJNA896729. All nitrifier genomes in this study were downloaded from the public NCBI RefSeq database (<https://www.ncbi.nlm.nih.gov/refseq/>).

Supplementary Information is available for this paper.

Correspondence and requests for materials should be addressed to Wei Qin (weiqin@ou.edu), Willm Martens-Habbena (w.martenshabbena@ufl.edu), and Mari Winkler (mwinkler@uw.edu).

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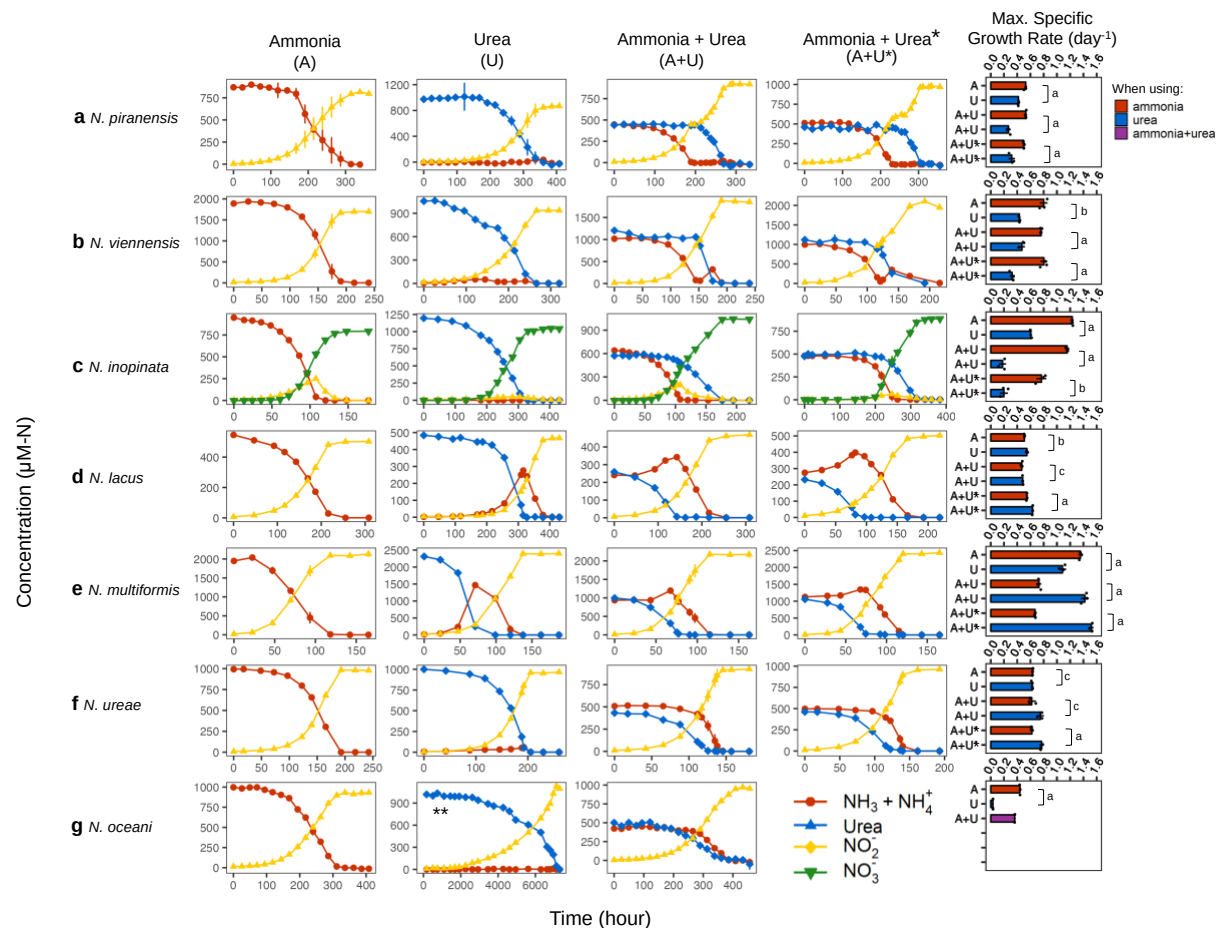
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Author Contributions: WQ, DAS, BBW, XM, WMH, and MKHW designed the experiments. WQ, SPW, YZ, EC, XL, JJ, XW, BA, and ZF performed the experiments and analyzed data with input from BW, HL, LH, QT, WWC, XS, MW, LN, and KAH. WQ, SPW, DAS, WMH, and MKHW wrote the manuscript, with contributions and approval from all other authors.

Competing Interests: The authors declare no conflicts of interest.

762 **Figure Legends**



763

764 **Fig. 1. Growth curves and maximum specific growth rate (μ_{max}) of (a) *N. piranensis* (marine**

765 **AOA), (b) *N. viennensis* (soil AOA), (c) *N. inopinata* (comammox), (d) *N. lacus* (*Nitrosospira***

766 **β -AOB), (e) *N. multiformis* (*Nitrosospira* β -AOB), (f) *N. ureae* (*Nitrosomonas* β -AOB), and (g)**

767 ***N. oceani* (γ -AOB). *Inoculation with 1% urea-grown culture at mid-exponential phase; the**

768 **other group of incubations on ammonia-urea mixtures were inoculated with 1% ammonia-grown**

769 **culture. Markers and error bars represent average and one standard deviation of biological**

770 **triplicates. The standard deviation is smaller than the marker if error bar is not visible. **Data**

771 **shown for *N. oceani*'s urea-only incubation from a single incubation without replicates. Urea-**

772 **only incubations inoculated with urea-grown culture were not conducted for *N. oceani* due to the**

773 **extremely slow growth rate. Letters indicate incubation with different N substrates: A =**

774 ammonia only; U = urea only; A+U = ammonia + urea. Column heights and error bars represent
775 average and one standard deviation of biological triplicates for specific growth rate
776 measurement. Lower case letters indicate the range of P values calculated from two-tailed t-test
777 pairing. a: $P < 0.01$, b: $0.01 < P < 0.05$, and c: $P > 0.05$. (See Supplementary Table 1 for all P
778 values).

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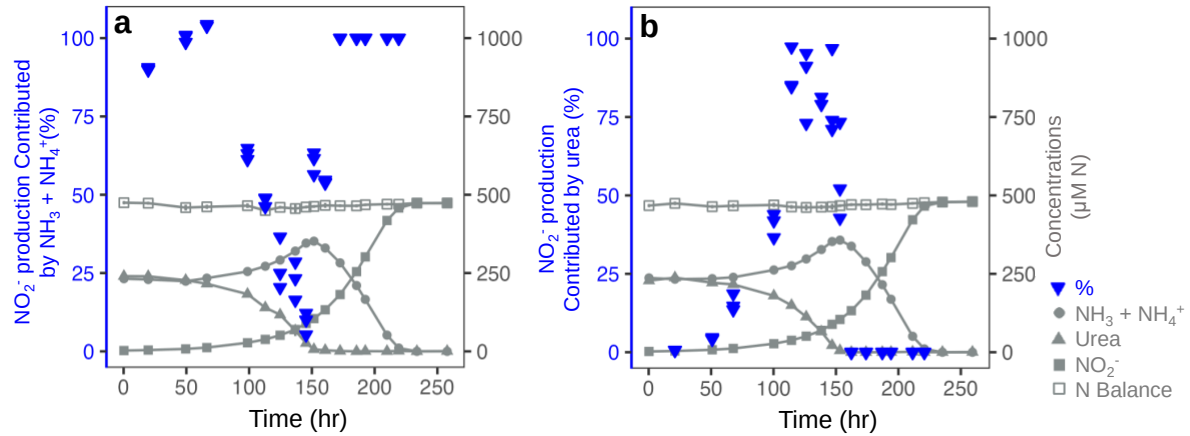


Fig. 2. ^{15}N stable isotope tracking the ammonia and urea utilization in *N. lacus*. The ^{15}N percentage of produced nitrite in the medium was measured for $^{15}\text{NH}_3$ tracer incubations (a) and ^{15}N -urea tracer incubations (b), in order to calculate the contribution of ammonia and urea to nitrite production. Total concentrations of N species are plotted in light grey on the secondary y axis for background reference.

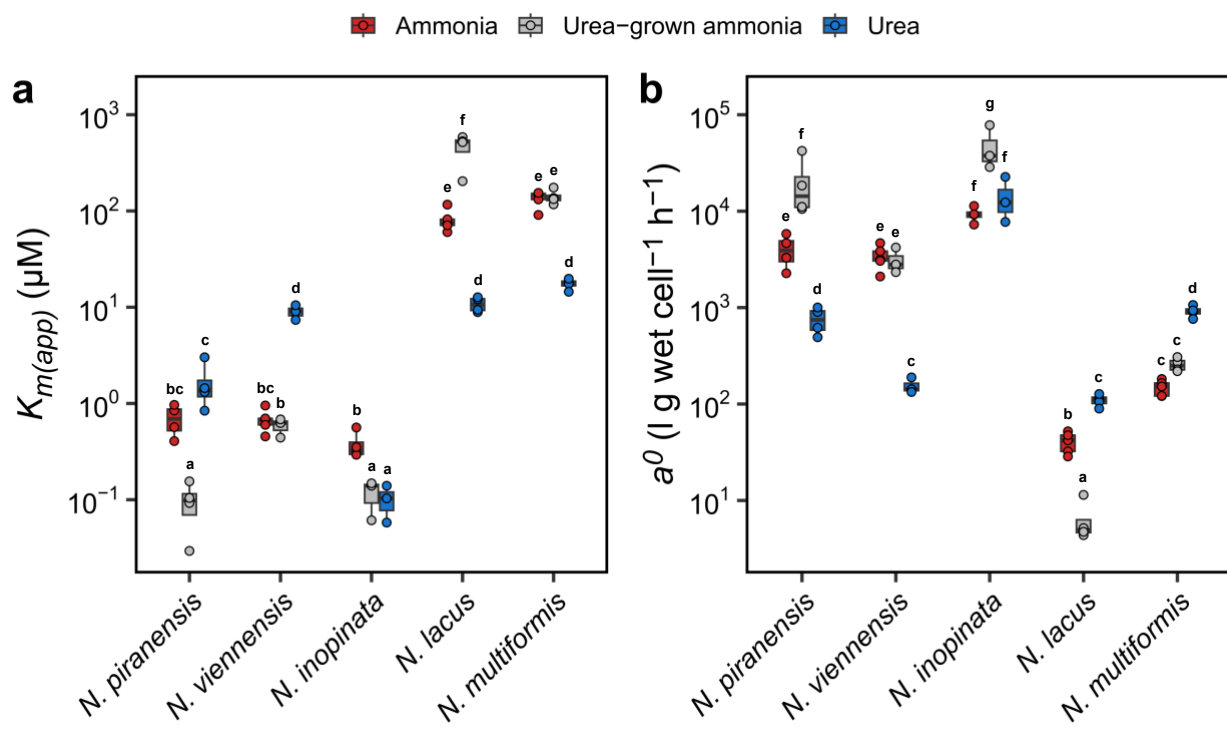


Fig. 3. Michaelis-Menten kinetic constants of ammonia- and urea-dependent oxidation. (a) Half saturation constant ($K_{m(app)}$), (b) specific substrate affinity (a°). $K_{m(app)}$ and a° for ammonia were determined both in ammonia- (red) and urea-grown cultures (grey), and $K_{m(app)}$ and a° for urea were determined in urea-grown species (blue). The upper limit, middle line, and lower limit of the boxplots indicate the 25th, 50th (median), and 75th percentile, respectively. Dots represent each replicate ($n = 3\text{--}5$ biologically independent experiments). Different lower-case letters denote statistically different groups based on Tukey's HSD after ANOVA comparison of means ($P < 0.05$). Data were $\log_{10}$ transformed to fulfill normal distribution, and Bonferroni correction was used to account for multiple comparisons.

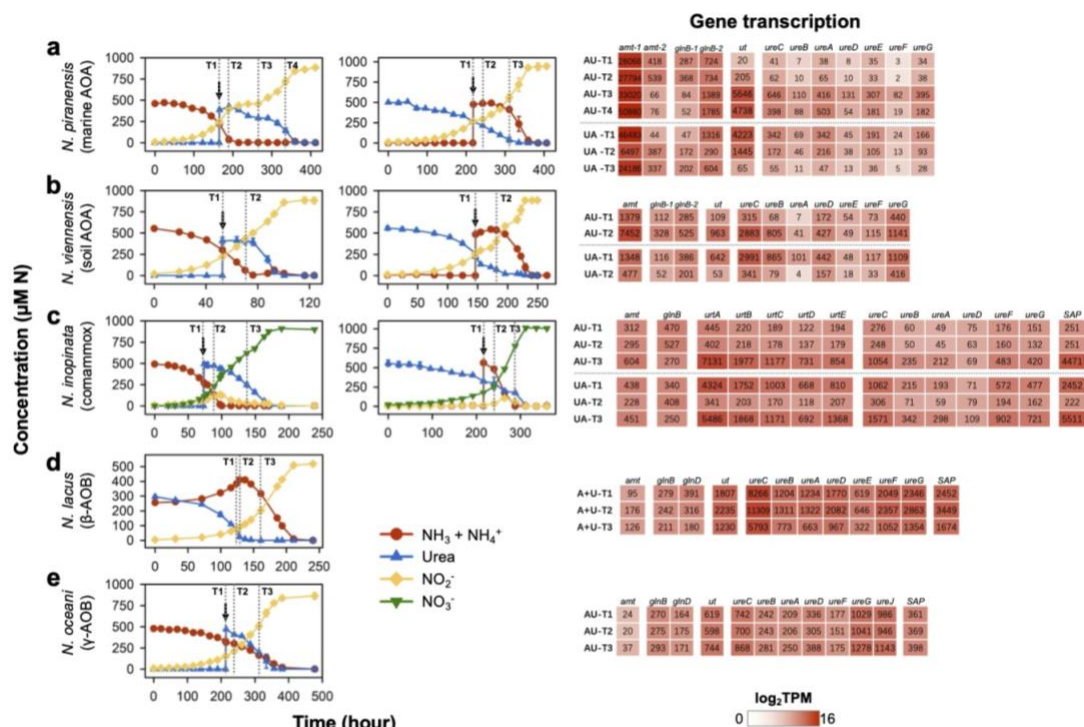


Fig. 4. Transcriptional response of AOM species to substrate switch from singe N source to a mixture of ammonia and urea. The transcriptional changes of ammonia and urea-utilizing genes were investigated 24 h after addition of an alternative N substrate to exponential phase ammonia-grown or urea-grown (a) *N. piranensis*, (b) *N. viennensis*, (c) *N. inopinata*, (d) *N. lacus*, and (e) *N. oceanii*. The growth data showing the times at which the alternative N substrate was added (arrows) and when cells were collected for transcriptomics analysis (dashed lines, T1–T4) are displayed in the left panels. T1 and T2 represent the sampling time points before the alternative N substrate addition and at ~24 hours after addition, respectively. AU = ammonia spiked urea; UA = urea spiked ammonia; A+U = ammonia+urea. Markers and error bars on growth curves represent the average and one standard deviation of biological triplicates. The transcription data for key ammonia and urea-utilizing genes are profiled in the right panels. The average transcripts per million (TPM) values for each gene are shown in colored boxes and the color scale bar represents the log₂(TPM). Gene full name and abbreviation pairs and source data are provided in Supplementary Table 3 and Source Data Table 1, respectively.

811

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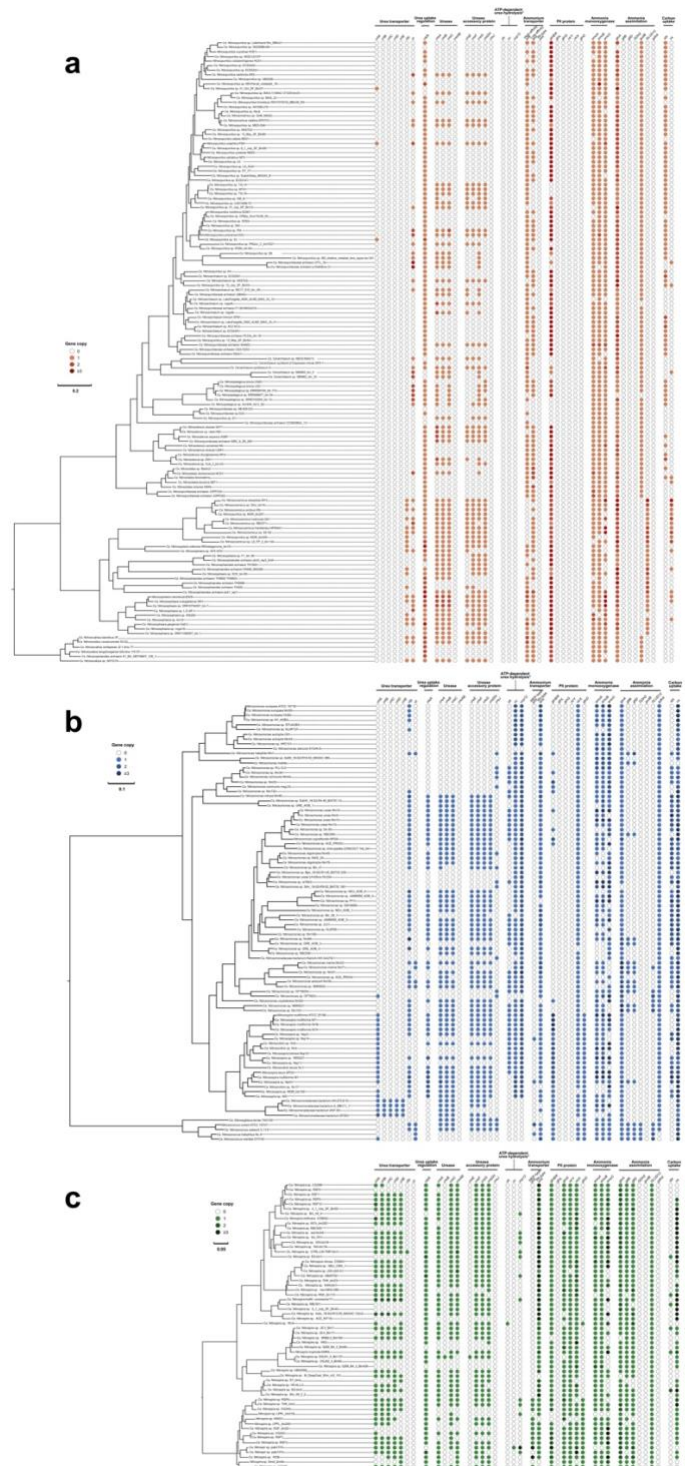
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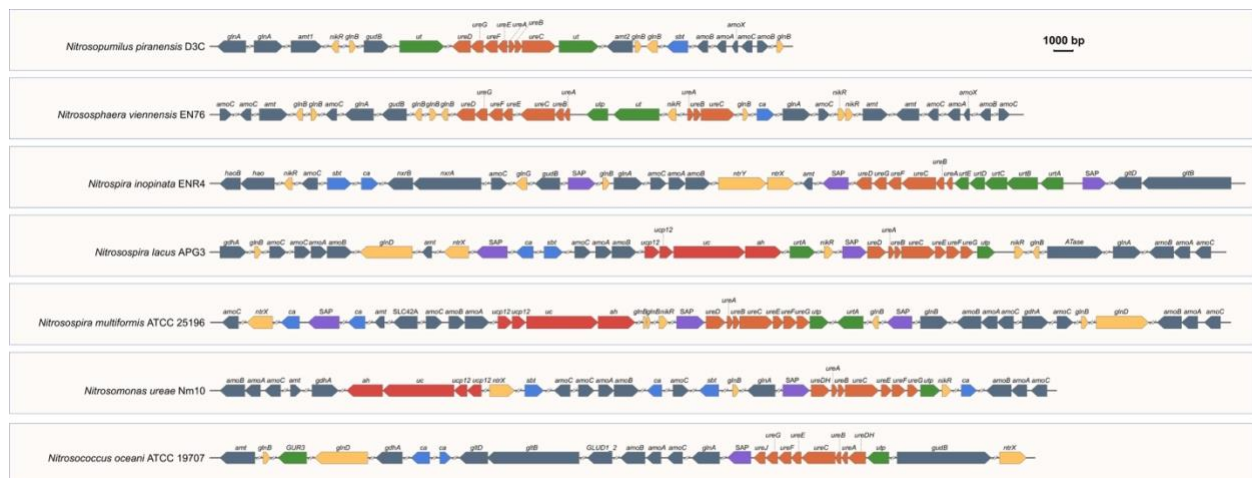
992 **Extended Figure Legend**



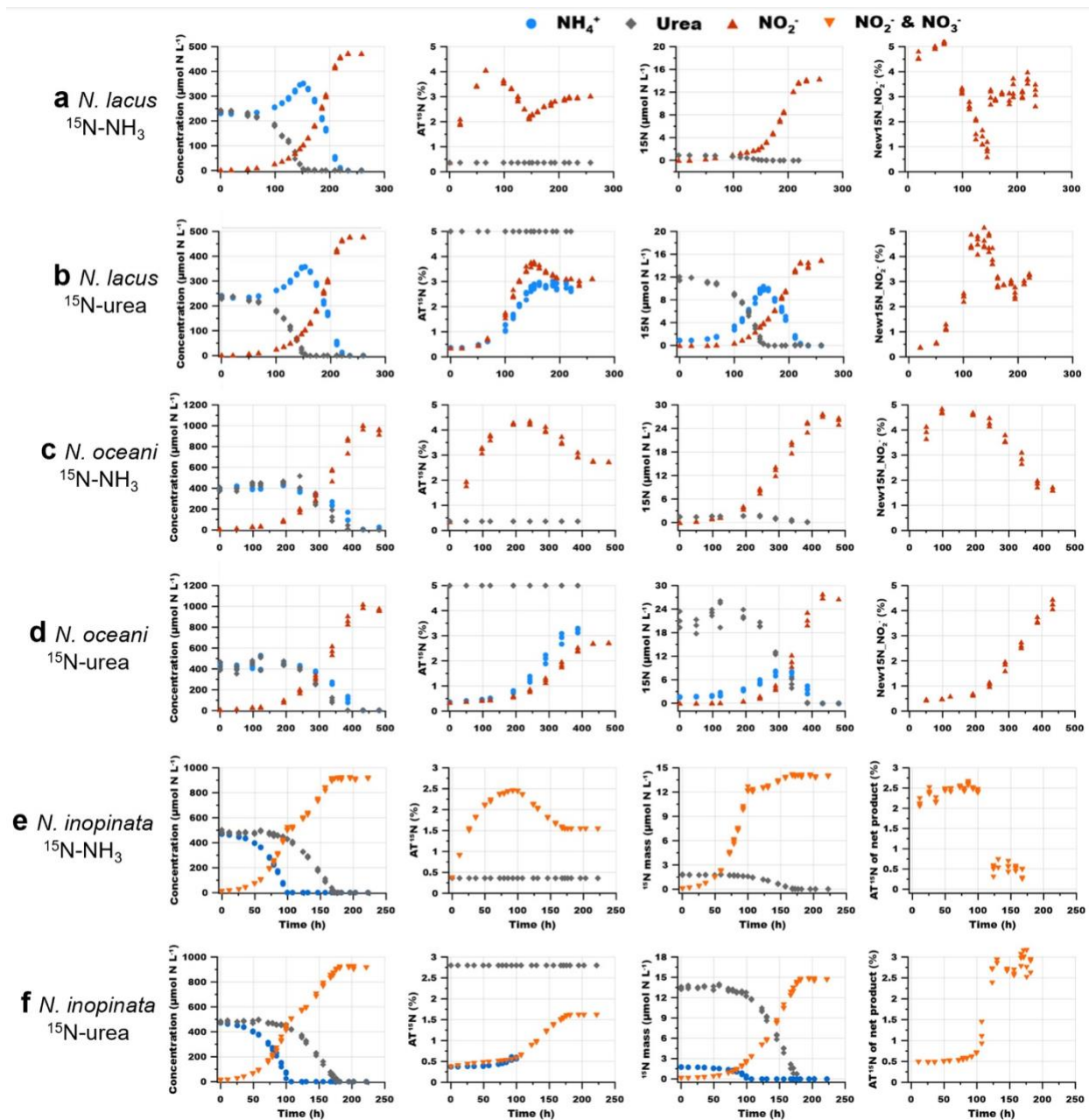
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994 **Extended Data Fig. 1 The phylogenetic pattern of ammonia and urea uptake and**
 995 **utilization genes in AOA (a), AOB (b), and comammox (c). The phylogenomic trees were**

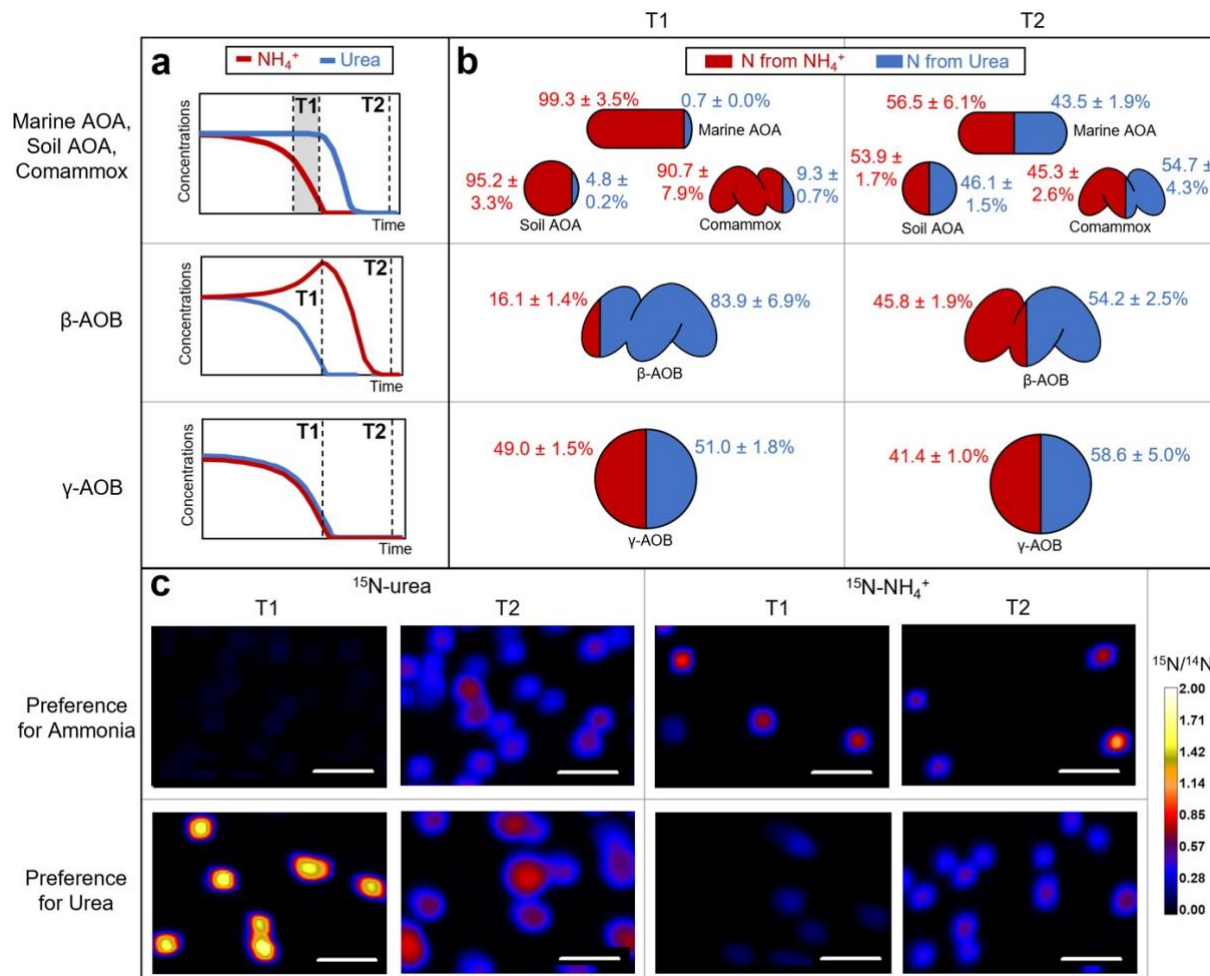
constructed based on concatenated sequences of 122 and 120 single-copy marker genes shared among archaea and bacteria, respectively⁷³. Confidence values are based on 1000 bootstrap replications. Scale bars represent 20%, 10%, and 5% estimated sequence divergence for AOA, AOB, and comammox phylogenomic trees, respectively. The estimated completeness of each genome is displayed next to the taxon name. The gene copy number is indicated by the darkness of the color in the cycles. *Note that the genes involved in the ATP-dependent urea hydrolysis pathway were transcribed at very low levels during urea consumption by β -AOB *N. lacus*, suggesting a different function of these conserved enzymes in β -AOB. *ut*: DUR3-type urea transporter; *urt*: ATP-dependent urea ABC transporter; *utp*: *yut*-type urea transporter; *nikR*: nickel-responsive transcriptional regulator; *ureA–C*: urease; *ureD–H, J*: urease accessory protein; *uc*: putative urea carboxylase; *ucp12*: putative urea carboxylase accessory protein; *amt*: Amt-type ammonium transporter; *Rh-type amt*, Rhesus-type ammonia transporter; *glnB/K*: nitrogen regulatory P_{II} protein; *glnL*: nitrogen regulation sensor histidine kinase GlnL; *glnG*: nitrogen regulation response regulator GlnG; *ntrX*: nitrogen regulation response regulator NtrX; *ntrY*: nitrogen regulation sensor histidine kinase NtrY; *glnD*: bifunctional P_{II} protein uridylyltransferase/uridylyl-removing enzyme; *amo*: ammonia monooxygenase; *gudB*: glutamate dehydrogenase; *GDH2*: glutamate dehydrogenase; *GLUD1_2*: glutamate dehydrogenase (NAD(P)⁺); *gdhA*: glutamate dehydrogenase (NADP⁺); *glt*: NADPH-dependent glutamate synthase; *glnA*: glutamine synthetase; *sbt*: bicarbonate transporter; *ca*: carbonic anhydrase; *SAP*: short chain amide/urea porin.



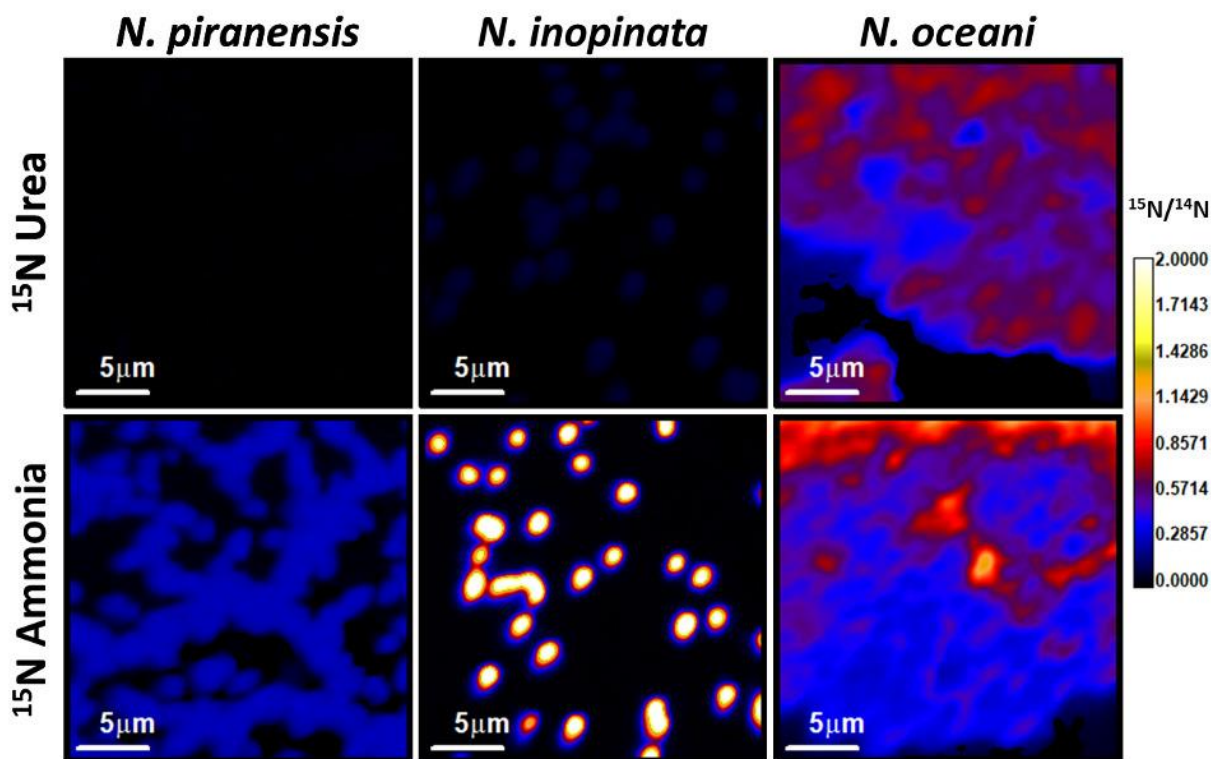
Extended Data Fig. 2 Schematic illustration of the genomic regions of the AOM species tested in this study containing genes for ammonia and urea uptake and utilization. Gene functions are indicated by color: Grey, ammonia oxidation and assimilation; green, urea transporters; purple, putative outer membrane short chain amide/urea porins; orange, urease and accessory proteins; yellow, nitrogen regulatory P_{II} proteins and Nickel-responsive regulator; blue, bicarbonate transporter and carbonic anhydrase; red, putative allophanate hydrolase, urea carboxylase, and accessory proteins. Genes are drawn to scale, and the scale bar corresponds to 1000 bp sequence length. Gene full name/abbreviation pairs are the same as shown in Extended Data Fig. 1 and Supplementary Table 3.



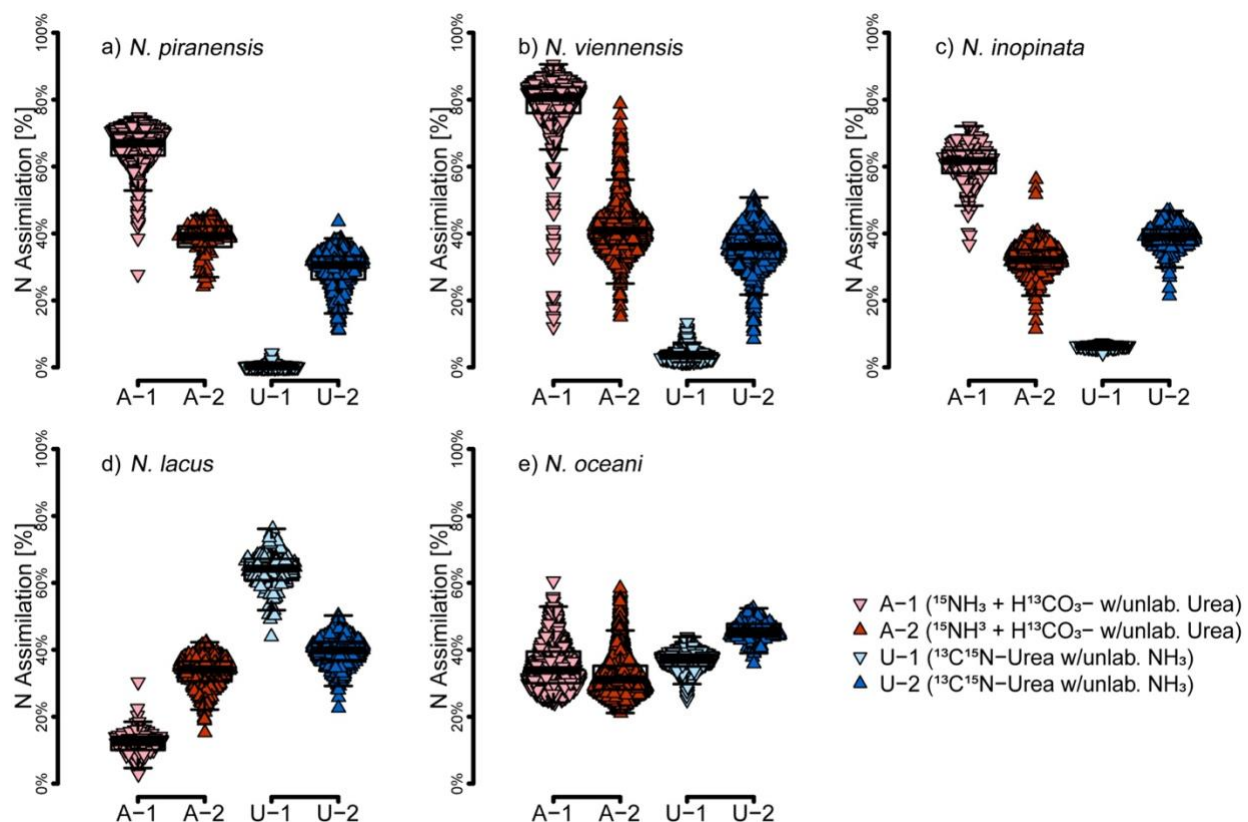
Extended Data Fig. 3 15N stable isotope tracking the nitrite/nitrate production and ammonia secretion in *N. lacus*, *N. oceani* and *N. inopinata*. Tracking nitrite/nitrate production and ammonia secretion for the incubations containing 15N-ammonia or 15N-urea. (a and b) *N. lacus*, (c and d) *N. oceani*, and (e and f) *N. inopinata* (n = 3 or 4 biologically independent incubations).



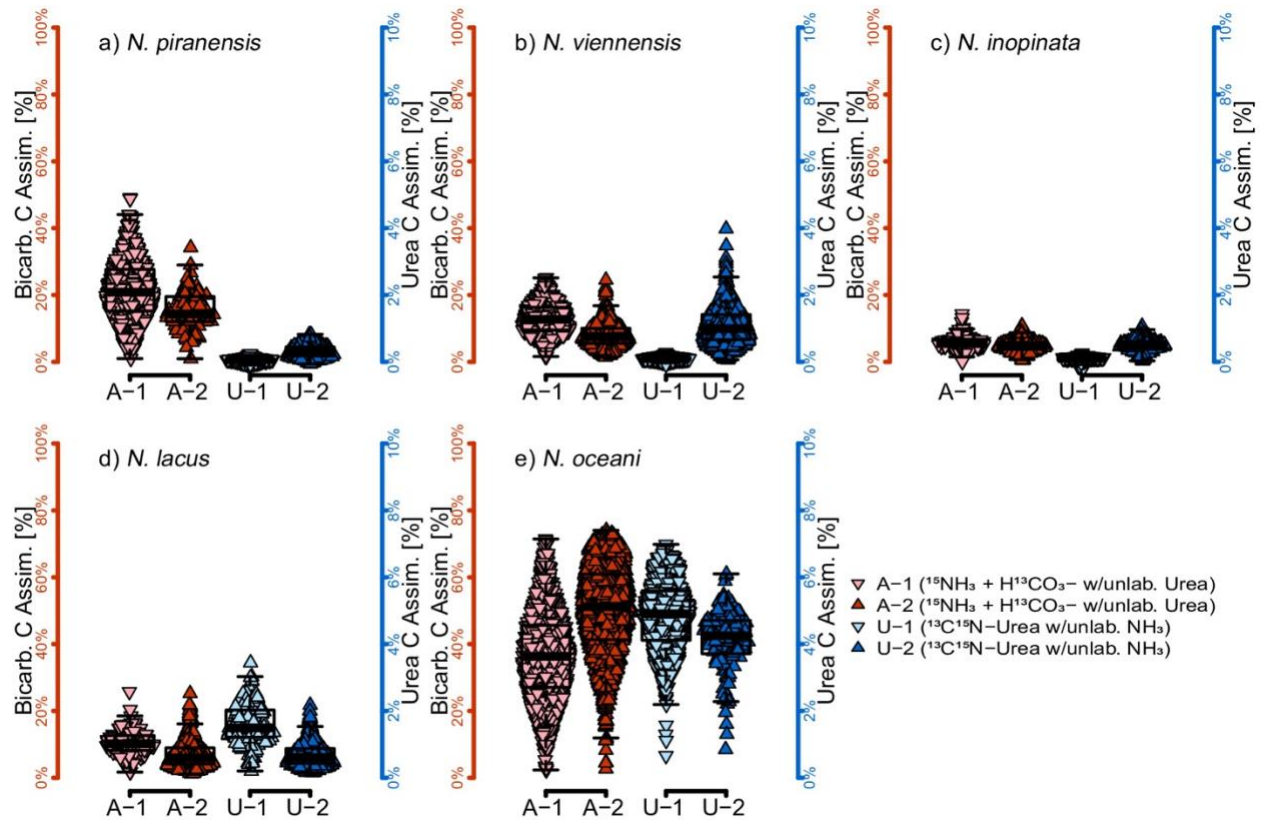
Extended Data Fig. 4 Summary of N assimilation results based on NanoSIMS. (a) Schematic representations of ammonia and urea consumption profiles and indication of time points for cell collection. (b) % new N incorporation (average $\pm$ standard deviation of all analyzed cells) from ammonia and urea. (c) Example NanoSIMS image (scale bar = 5 μm) where top row shows preference for ammonia (images of the soil AOA *N. viennensis* cells) and bottom shows preference for urea (images of the β -AOB *N. lacus* cells). Over 15 NanoSIMS images were collected for each AOM species at each time point. See Supplementary Figs. 4 and Extended Data Fig. 5 for growth curves with the time points indicated for each organism and the NanoSIMS images for other investigated AOM species, respectively. The NanoSIMS analysis software output data, which were processed to calculate % incorporation from substrates, are provided in Source Data Table 2.



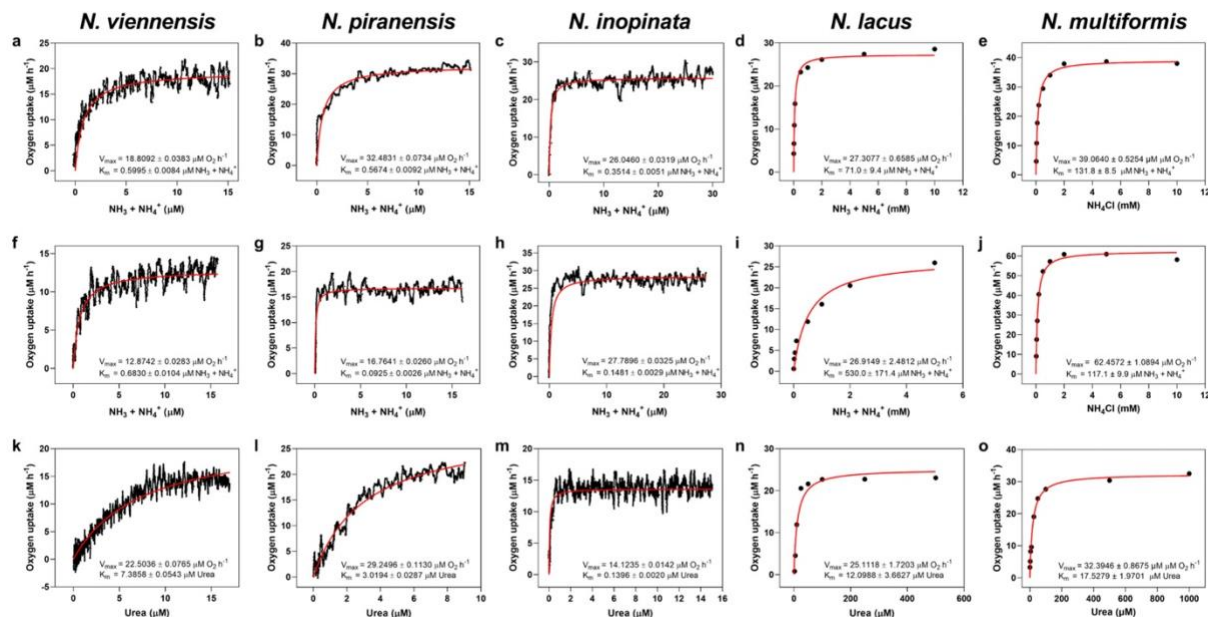
Extended Data Fig. 5 Example NanoSIMS images for the marine AOA *N. piranensis*, comammox *N. inopinata*, and γ -AOB *N. oceani*. Top and bottom rows represent the NanoSIMS images taken at T1 for labeled urea and labeled ammonia incubations, respectively. Over 15 NanoSIMS images were collected for each AOM species at each time point. The NanoSIMS analysis software output data, which were processed to calculate % incorporation from substrates, are provided in Source Data Table 2.



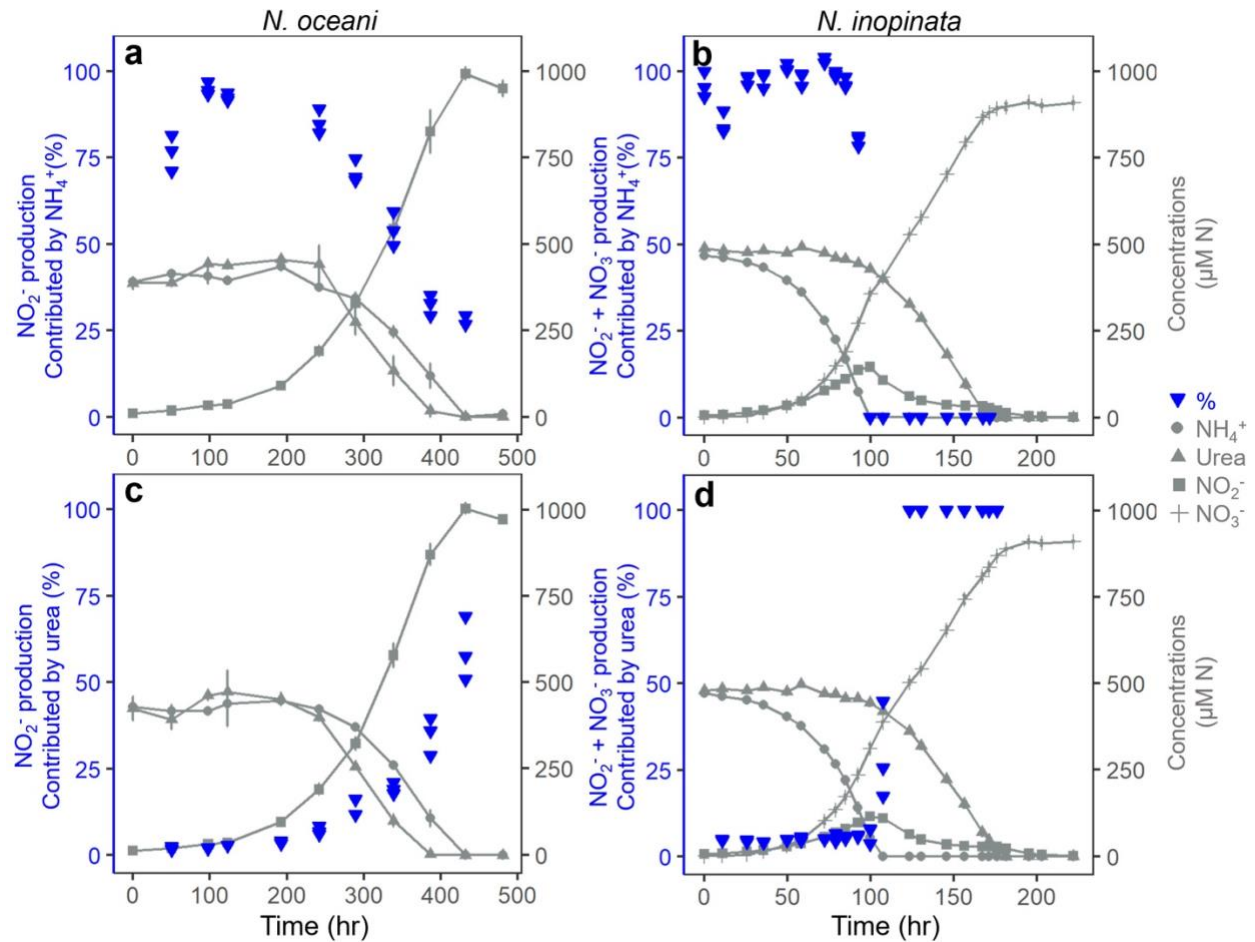
Extended Data Fig. 6 Percent of N incorporation from ammonia-N or urea-N determined by nanoSIMS for single cells of (a) *N. piranensis* (marine AOA), (b) *N. viennensis* (soil AOA), (c) *N. inopinata* (comammox), (d) *N. lacus* (β -AOB), and (e) *N. oceani* (γ -AOB). N-incorporation data were obtained in multiple filters harvesting cells from biological triplicate incubations ($n = 3$). The box plots represent the median as well as the 25% and 75% interquartile ranges. The whiskers represent $1.5 \times$ the interquartile ranges.



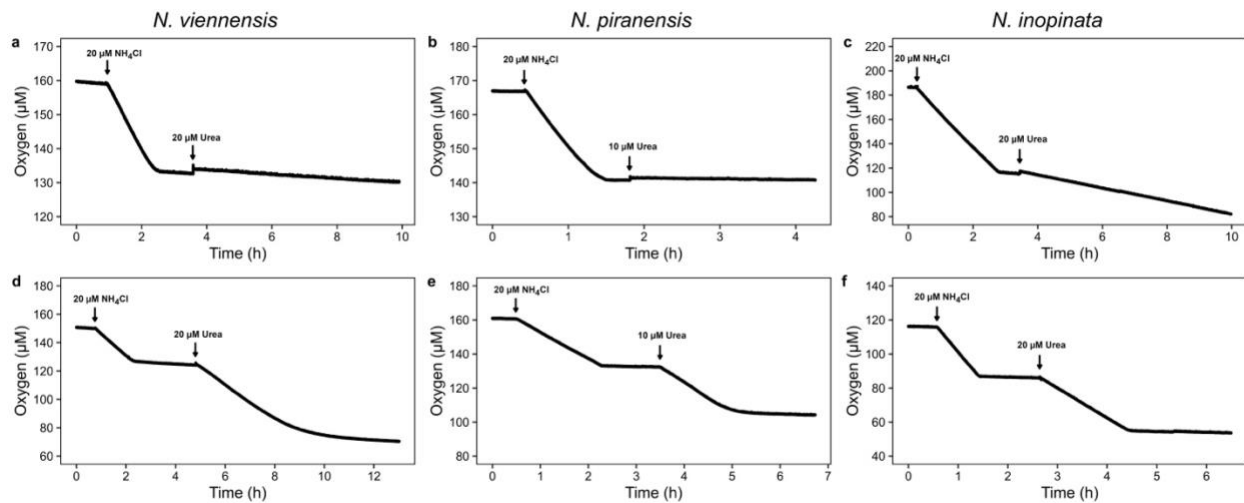
Extended Data Fig. 7 Percent of C incorporation from bicarbonate-C or urea-C determined by NanoSIMS for single cells of (a) *N. piranensis* (marine AOA), (b) *N. viennensis* (soil AOA), (c) *N. inopinata* (comammox), (d) *N. lacus* (β -AOB), and (e) *N. oceanii* (γ -AOB). Note that Y-axes are in different scales for bicarbonate-C (red) and urea-C (blue) incorporation. C incorporation data were obtained in multiple filters harvesting cells from biological triplicate incubations ($n = 3$). The box plots represent the median as well as the 25% and 75% interquartile ranges. The whiskers represent $1.5\times$ the interquartile ranges.



Extended Data Fig. 8 Ammonia- or urea-dependent substrate oxidation kinetics of *N. viennensis*, *N. piranensis*, *N. inopinata*, *N. lacus*, and *N. multiformis*. a–e, Ammonia-grown strains, and f–o, Urea-grown strains. Each graph represents a single trace of experimental replicates. Oxygen uptake rates were determined by a microrespirometry (MR) measurement upon substrate addition. Kinetic constants were obtained by fitting a Michaelis-Menten equation (red line) to total ammonia or urea and oxidation rates.



Extended Data Fig. 9 ^{15}N stable isotope tracking the ammonia and urea utilization in *N. oceanii* and *N. inopinata*. The ^{15}N percentage of accumulated nitrite in the medium was measured for ^{15}N - NH_3 treated incubations (a and b) and ^{15}N -urea treated incubations (c and d) and the contribution of ammonia and urea to nitrite production was calculated for (a, c) *N. oceanii* and to nitrite and nitrate production for (b, d) *N. inopinata* ($n = 3$ biologically independent incubations). Error bars represent the standard deviations of the biological triplicate incubations. Total concentrations of N species are plotted in light grey on the secondary y axis for background reference.



Extended Data Fig. 10 Instantaneous measurement of oxygen uptake with *N. viennensis*, *N. piranensis*, and *N. inopinata*. a–c, Ammonia-grown strains, and d–f, urea-grown strains. Arrows: the injection of ammonia or urea into the chamber via an injection port using a 2 mL syringe.

Supplementary Discussion

Stable isotope tracing of the cellular fate of N in *N. oceani* and *N. inopinata*

The ^{15}N -labelling approach was used to track the fate of N species in the growth experiments of *N. oceani* and *N. inopinata* (Extended Data Figs. 3c–f, 9, Supplementary Fig. 7). As indicated by the unlabeled growth experiments (Fig. 1g), the simultaneous consumption of ammonia and urea by the γ -AOB *N. oceani* was examined using the same ^{15}N -labeling approach as described for *N. lacus* labeling experiments (Fig. 2). Cultures inoculated with ammonia-preadapted *N. oceani* cells immediately converted $^{15}\text{NH}_3$ to $^{15}\text{NO}_2^-$ during the lag phase (0–100 h) (Extended Data Fig. 9a). In the parallel experiment, very little $^{15}\text{NO}_2^-$ derived from ^{15}N -urea during lag phase (Extended Data Fig. 9c). The percentage of $^{15}\text{NO}_2^-$ originating from ^{15}N -urea then gradually increased from ~3% to ~60% during exponential growth (Extended Data Fig. 9c), which was coupled with a continuous decrease in the contribution from NH_3 , as shown in the parallel experiment tracing $^{15}\text{NH}_3$ to $^{15}\text{NO}_2^-$ production throughout the remaining growth period (Extended Data Fig. 9a). Although no net accumulation of ammonia was observed in unlabeled growth experiments, a substantial amount of $^{15}\text{NH}_3$ (~55%) was released into the medium during ^{15}N -urea hydrolysis (Extended Data Fig. 3d, Supplementary Fig. 7). The release of $^{15}\text{NH}_3$ was closely correlated with $^{15}\text{NO}_2^-$ production ($R^2 = 0.99$, $P < 0.01$) in the ^{15}N -urea labeling experiments (Supplementary Fig. 7). Thus, about 45% of the ammonia released to the medium during urea hydrolysis was rapidly oxidized to nitrite.

Unlike the AOA species, comammox *N. inopinata* showed a weaker repression of urea utilization and started to use urea before ammonia was fully exhausted (Fig. 1c). In incubations with added ^{15}N -urea, the urea-derived ^{15}N atoms appeared in the extracellular pools of nitrite, nitrate, and ammonia ~75 h after incubation, showing that *N. inopinata* turned on urea utilization significantly prior to ammonia depletion (at ~100 h) and suggesting a mechanism for anticipatory transition to urea consumption (Extended Data Fig. 3f). This transition was

associated with a reduction in growth rate when consuming ammonia in the mixed N species medium compared to growth in the ammonia-only medium ($P < 0.001$) (Fig. 1c), presumably because of the metabolic burden of expressing urea utilization genes while ammonia was still present. After the complete shift to urea consumption (~100 h), the secreted $^{15}\text{NH}_3$ decreased to below the level of detection (1 nM) (Extended Data Fig. 3f).

Preferred nitrogen substrate assimilation in AOA, comammox, and γ -AOB species

Similar to β -AOB *N. lacus*, we conducted a parallel set of labeling experiments with marine AOA *N. piranensis*, soil AOA *N. viennensis*, and comammox *N. inopinata* (Extended Data Fig. 4a), and examined their ammonia- and urea-N assimilation at single-cell resolution using NanoSIMS. In contrast to *N. lacus*, they all preferentially assimilated N from ammonia at T1 with $99.3 \pm 3.5\%$, $95.2 \pm 3.3\%$, and $90.7 \pm 7.9\%$ new N incorporation from ammonia and $0.7 \pm 0.0\%$, $4.8 \pm 0.2\%$, and $9.3 \pm 0.7\%$ from urea for *N. piranensis*, *N. viennensis*, and *N. inopinata*, respectively (Extended Data Fig. 4b, c). At T2, new N incorporation from urea and ammonia was roughly equivalent at 43.5–56.5% (Extended Data Fig. 4b, c). For the third type of behavior shown by the γ -AOB *N. oceanii*, we found similar new N incorporation from urea and ammonia in late exponential phase (49.0–51.0%) (Extended Data Fig. 4b, c), showing that γ -AOB co-assimilated ammonia and urea.

Distinct responses of AOA and comammox species after transitioning to consumption of the secondary N substrate

Although the characterized AOA and comammox species all prefer ammonia as their primary energy and N source, there was large variation in their adaptive transition from ammonia to urea metabolism. The marine AOA *N. piranensis* demonstrated a pronounced reduction in growth rate upon ammonia depletion, while the soil AOA *N. viennensis* showed negligible lag when grown on a mixture of ammonia and urea (Fig. 1a, b). The comammox species *N. inopinata* initiated urea utilization ~25 hours before ammonia exhaustion (Extended Data Fig. 3f), possibly

reflecting an adaptive mechanism to use the alternative substrate before ammonia depletion but at the cost of the added biosynthetic burden slowing growth before full transition (Fig. 1c). In contrast, both characterized marine and soil AOA species repressed use of urea, growing on ammonia at comparable rates in the presence or absence of urea before transitioning to urea consumption (Fig. 1a, b).

In addition, we characterized the response of AOA and comammox species to the switch from a single N substrate to ammonia-urea mixture. The addition of urea to the AOA *N. piranensis* and *N. viennensis* did not interrupt their growth on ammonia. Both organisms repressed the consumption of urea until ammonia was depleted (Fig. 4a, b). However, they differed in the growth kinetics of substrate transition. *N. viennensis* sustained active growth during transition whereas *N. piranensis* exhibited a significant lag (Fig. 4a, b). With the alternate order of addition, growth of both species slowed upon ammonia addition and active ammonia consumption was delayed until near depletion of urea. (Fig. 4a, b). Unlike the AOA species, the comammox *N. inopinata* showed much weaker repression of urea consumption in the presence of ammonia and immediately consumed urea upon its addition to ammonia-growing cultures (Fig. 4c).

We also characterized the oxygen uptake activity of AOA and comammox species after addition of low concentrations of ammonia and urea using microrespirometry (Extended Data Fig. 10). Following multiple subculturing in ammonia or urea-only media, we observed instantaneous oxygen consumption upon addition of low concentrations of ammonia (20–40 μ M) or urea (10–20 μ M) to urea-grown *N. piranensis* and *N. viennensis* cells at early stationary phase (Extended Data Fig. 10d, e). No discernible stimulation of O₂ uptake was observed for multiple hours when 10 μ M urea was added to ammonia-grown cells immediately following their depletion of ammonia from the medium (Extended Data Fig. 10a, b). This further indicated that these AOA species repress urea utilization while growing on ammonia and that a significant period of adaptation is required before their transition to growth on urea. Repression of growth on urea is

also consistent with AOA having evolved as specialists in competition for ammonia in mixed-nutrient environments (Figs. 3 and 4). In contrast, both ammonia- and urea-adapted comammox *N. inopinata* responded quickly to addition of either substrate (Extended Data Fig. 10c, f). Thus, comammox appeared to function as a generalist, having relatively high affinity for both substrates but less stringent regulation. Previous studies have demonstrated a remarkable metabolic versatility among *Nitrospira* species¹. Thus, it would also be of interest to investigate whether *N. inopinata* utilizes other organic N sources and if it possesses high affinity transporters for these substrates as well. Such unique adaptive features could provide *N. inopinata*-like comammox species with a competitive advantage in environments with variable supplies of inorganic and organic N substrates².

P_{II} proteins and their modifying enzymes control the selective N source uptake regulation in AOM

Our combined analyses clearly implicated GlnB-type P_{II} proteins in controlling the uptake of extracellular N species through interaction with AMT. This family of proteins have well characterized functions in modulating the uptake of ammonia used for assimilation^{34,35} but little is known about how they function in organisms that use ammonia for both anabolic synthesis and catabolic energy generation. In well characterized heterotrophs, the transport of ammonia by the AMT can be blocked by binding to the GlnB protein. Binding to GlnB is controlled by its post-translational modification by a second regulatory protein (GlnD) in response to variations in intracellular glutamine and 2-oxoglutarate levels³⁻⁷. Although GlnB-type P_{II} proteins are common in all major lineages of AOM, they are not always associated with GlnD. A homolog of GlnD could not be identified in AOA and is variably distributed among AOB (Extended Data Fig. 1a, b). It is present in *Nitrosospira* β -AOB and *Nitrosococcus* γ -AOB but absent in *Nitrosomonas* β -AOB (Extended Data Fig. 1b).

N. lacus carries the full canonical set of N uptake regulatory genes. The absence of any apparent lag in the growth of *N. lacus* with depletion of urea in the mixed N medium, and blocked transport of ammonia before urea depletion, were consistent with GlnB-type P_{II} protein control of ammonia uptake (Figs. 1d and 2). Since this P_{II} protein is the only known substrate for GlnD⁷, and *glnD* was transcribed at significantly higher levels ($P = 6.57 \times 10^{-15}$) during ammonia uptake repression (Fig. 4d), our data strongly suggest GlnB-type P_{II} protein was post-translationally modified by GlnD via reversible uridylylation and deuridylylation. Although the activity of GlnD is known to be controlled by intracellular glutamine levels as a measure of cellular N sufficiency^{6,7}, the absence of any change in growth rate with transition from urea to growth on ammonia suggested that ammonia was never limiting. Nitrogen sufficiency throughout growth was also indicated by relatively unaltered transcription of genes coding for ammonia assimilation enzymes, such as glutamate dehydrogenase (GDH), glutamine synthetase (GS), and the GS modifying enzyme glutamine synthetase adenylyltransferase (GlnE) (Supplementary Table 3). Assuming that levels of glutamine and other known intracellular metabolites did not signal N deficiency, GlnD might be responding to the availability of an effector molecule other than glutamine, possibly urea, in *Nitrosospira* β-AOB species.

Unlike *N. lacus*, the AOA species repressed urea uptake and utilization when cells were using ammonia and demonstrated a more complex role of P_{II} proteins. The *N. piranensis* genome harbors two copies of DUR3-type urea transporters (NPIRD3C_1395 and NPIRD3C_1384), and only NPIRD3C_1395 was highly transcribed on urea, suggesting it was mainly responsible for urea uptake. *N. piranensis* also encodes two ammonia transporter genes, *amt1* and *amt2*, suggested to have high and low affinity for ammonia, respectively, based on their similarity to transporters characterized in *Nitrosopumilus maritimus*⁸⁻¹⁰. The *amt1* gene was constitutively transcribed under all growth conditions (Fig. 4a). Transcription of the lower affinity *amt2* (NPIRD3C_1670), transcribed at much lower levels than *amt1* (Fig. 4a), was elevated in the presence of ammonia (AU-T1 and T2) and in response to ammonia addition to cultures growing

on urea (UA-T2 vs. T1), and depressed upon ammonia depletion (AU-T3 and T4). The differential response of *amt2* relative to *amt1* suggests *amt2* may have additional functions other than ammonia uptake.

Two *glnB*-like genes (*glnB1* and *glnB2*, NPIRD3C_1671 and NPIRD3C_1673) flank the downstream region of *amt2* (Extended Data Fig. 2). Intriguingly, *glnB1* (NPIRD3C_1671) was significantly downregulated ($P = 3.5 \times 10^{-20}$) and the high affinity *amt1* significantly upregulated after switching from ammonia to urea consumption and vice versa upon ammonia introduction into urea-grown cells (Fig. 4a). The second *glnB*-like gene (NPIRD3C_2171) showed the opposite transcription pattern (Fig. 4a). Although constitutively transcribed at relatively high levels in *N. piranensis*, it was significantly upregulated ($P = 9.3 \times 10^{-4}$) with transition from ammonia to urea (AU-T3 vs. T2) and significantly repressed ($P = 2.7 \times 10^{-56}$) in response to the addition of ammonia (UA-T2 vs. T1). These results strongly suggest a role for these P_{II} proteins in coordinating the activities of *amt1* and *amt2* in ammonia transport and the sensing of cellular N status.

Differential transcription of two putative *glnB* genes was also observed in the soil AOA *N. viennensis* (NVIE_013340, NVIE_014550) in response to ammonia and urea additions. Their transcription levels were significantly higher ($P < 0.01$) following urea spike and were significantly lower ($P < 0.01$) after ammonia introduction (Fig. 4b).

Two *glnB* genes (NPIRD3C_1671 and NPIRD3C_2171) in marine AOA *N. piranensis* showed opposite patterns of elevated transcription in response to extracellular ammonia and urea availabilities (Fig. 4a), suggesting they may carry out potentially complementary regulatory functions to cooperatively control influx of ammonia and urea in marine AOA. A range of different transporters have been recently identified as new P_{II} protein targets, including urea transporters¹¹. Thus, it is tempting to speculate that GlnB in AOA can also interact with urea transporters to block the uptake of external urea in the presence of ammonia. Neither *glnB* nor

glnD gene was differentially transcribed in γ -AOB *N. oceanii* (Fig. 4e), which is consistent with the lack of N source repression for this species when grown in a mixture of ammonia and urea. No homolog of *glnB* or *glnD* was identified in most *Nitrosomonas* β -AOB species genomes (Extended Data Fig. 1b), and *Nitrosomonas ureae* lacks the gene encoding GlnD, which may in part explain the weaker repression of ammonia utilization by *Nitrosomonas ureae* compared to *Nitrospira* species (Fig. 1f).

Transcriptional changes and transcription control of urea hydrolysis pathway genes in *N. lacus*

Urea hydrolysis genes (EBAPG3_007765–EBAPG3_007795) were constitutively transcribed in *N. lacus* across all growth phases. Transcripts of the *ureC* gene (EBAPG3_007780) were among the ten-most abundant in the transcriptome of *N. lacus* before urea exhaustion (Supplementary Tables 3 and Source Data Table 1). Although significantly downregulated upon urea depletion ($P = 2.8 \times 10^{-17}$) (A+U T3 vs. T2), transcript abundance remained elevated after switching to using ammonia (Fig. 4d).

Regulation of the urease operon is almost certainly controlled by a nickel-responsive regulator (NikR, EBAPG3_007745) located immediately upstream of the urea utilization operon in *N. lacus* (Extended Data Fig. 2), which has been shown to control transcription of the *Helicobacter pylori* urease¹². Its transcript abundance decreased ($P = 1.7 \times 10^{-239}$) more than 46-fold upon urea depletion (A+U-T3 vs. T2) (Supplementary Table 3).

N. lacus encodes enzymes implicated in ATP-dependent urea hydrolysis in other β -AOB species, urea carboxylase and allophanate hydrolase (Extended Data Figs. 1b and 2). However, their low transcription relative to the urease genes during urea consumption by *N. lacus* suggests different functions for these conserved enzymes in β -AOB (Supplementary Table 3).

Limited assimilation of urea-derived C by the tested AOM species

The direct assimilation of urea-derived C by AOM at the single-cell level was confirmed via NanoSIMS analysis, and we compared these incorporation rates to autotrophic C fixation rates (Extended Data Fig. 7). Archaeal and bacterial ammonia oxidizers were grown in ammonia-urea media with either dual-labeled $^{13}\text{C}^{15}\text{N}$ -urea (and unlabeled NH_3 and bicarbonate) or ^{15}N - NH_3 and ^{13}C -bicarbonate (and unlabeled urea). The marine AOA *N. piranensis* incorporated 15.3–21.3% of its total cellular C from bicarbonate but barely incorporated C from urea (0.1–0.3%) across different growth phases (Extended Data Fig. 7a). For the soil AOA *N. viennensis* and the comammox *N. inopinata*, hardly any incorporation of urea-C ($\sim 0.1\%$) was observed when grown on ammonia (T1), and C incorporation from urea increased to 0.5–1.1% after urea depletion (T2) (Extended Data Fig. 7b, c). Consistently, at T2, C incorporation from bicarbonate decreased from $13 \pm 4.5\%$ and $5.9 \pm 2.0\%$ to $8.0 \pm 3.5\%$ and $4.9 \pm 1.8\%$ for *N. viennensis* and *N. inopinata*, respectively (Extended Data Fig. 7b, c). The β -AOB *N. lacus*, which preferentially used urea-N over ammonia, incorporated $1.6 \pm 0.6\%$ C from urea at T1 (Extended Data Fig. 7d), which was less than from bicarbonate ($10.5 \pm 0.9\%$ C). The C incorporation from urea decreased to $0.7 \pm 0.3\%$ at T2 (Extended Data Fig. 7d). The percentage of urea-C incorporation remained relatively unchanged for the γ -AOB *N. oceani* across different growth phases (4.1–4.8%) (Extended Data Fig. 7e). Taken together, although some characterized AOM species were able to use urea-derived C to supplement their C requirement in the presence of bicarbonate and CO_2 , they mainly assimilated urea-derived N when grown on urea (Extended Data Figs. 4–7) and urea was not a significant source of cellular carbon.

Potential factors influencing kinetic analysis of AOM species

We do notice some differences among the kinetic data for ammonia oxidation particularly in *N. viennensis* between our study and a previous report¹³. Although we can only speculate about the reasons, strains with high specific affinities have been found sensitive to variations in temperature, laboratory handling, and substrate depletion¹⁴. The higher specific affinities for ammonia of *N. viennensis* (Fig. 3b and Supplementary Table 2) reported here may reflect a more

realistic estimate of the maximum specific affinities attainable by this species. Short-term kinetics experiments on whole cells can furthermore be prone to accumulation of intermediates and result in varying respiratory activities¹⁴. To avoid such metabolic imbalances as much as possible, we grew all cultures at sufficient O₂ to avoid O₂ limitation in late exponential phase and associated possible changes in expression of respiratory complexes. Kinetic properties of all strains were investigated without harvesting of cell material. Only in strains where near steady-state metabolic rates with 20 μ M ammonia addition were too low and $K_{m(app)}$ above the range measurable in individual incubations, measurements were conducted by multiple discrete substrate additions with increasing concentrations.

Function of Rhesus-type ammonia transporter in AOM species

Our study indicates that both ammonia assimilation and oxidation are transport-dependent. Although the *rh50 amt* knockout mutant of *Nitrosomonas europaea* can still oxidize ammonia when supplemented with high ammonia concentrations (≥ 1 mM)¹⁵, it is possible that ammonia (NH₃) diffusion alone was likely sufficient to support ammonia oxidation and assimilation for growth in the mutants under conditions of the relatively high ammonia concentrations tested. Indeed, previous studies of ammonia assimilation by *E. coli* have revealed that active ammonia transport via its ammonia transporter was essential for growth when ammonia concentrations were below ~ 20 μ M¹⁶. At higher ammonia concentrations, passive membrane diffusion becomes a major source of ammonia for biosynthesis in *E. coli*¹⁶. Considering that $\sim 99\%$ of the consumed ammonia was oxidized, while only $\sim 1\%$ was assimilated by *N. europaea*, it has been proposed that ammonia uptake capacity is regulated based on the demands of energy generation, rather than N assimilation¹⁷.

Both AOB and clade A comammox species encode Rh50-type AMTs, however, their characterized species have shown distinct affinities for ammonia¹³. While the Rh50-type AMT of *N. europaea* has been previously characterized as a low-affinity AMT¹⁸, recent studies have

indicated that even minor alterations to a twin histidine motif of AMTs can increase their affinity for ammonium by more than 10-fold¹⁹. Also, given that ammonia transport is essential for anabolism, the presence of a low-affinity AMT in *N. inopinata* appears incompatible with its oligotrophic lifestyle²⁰. Thus, direct biochemical characterization of the Rh50-type transporter of *N. inopinata* would be an intriguing next step.

Urea oxidation in marine environments

Previous field studies have reported high urea oxidation rates in marine environments, however, whether these transformations are solely attributed to a single group of organisms or involve a consortium of urea-hydrolyzing microbe and AOM species remains unresolved²¹. Incubation experiments in the Gulf of Mexico using urea isotope labeling suggest that marine AOA can directly oxidize a significant portion of urea-N without cross-feeding interactions²². However, it should be noted that in incubations amended with unlabeled ammonia, the rates of ¹⁵N-nitrite production (1.9-16.9 nM/day) were considerably lower than in those without added ¹⁴N-ammonia (24.5-58.7 nM/day), indicating that the presence of ammonia limited urea utilization by marine AOA²². In addition, Tolar *et al.* measured the release of ¹⁵N-ammonia in seawater samples amended with ¹⁵N-labeled urea to investigate the contribution of urea hydrolysis to the ammonia pool²³. They found that ~20% of the ¹⁵N-label from urea was converted into ¹⁵N-labeled ammonia in coastal and open ocean waters, suggesting that at least ~20% of free ammonia released from urea hydrolysis can fuel other ammonia oxidation or assimilation processes²³. Moreover, reports of the absence of *ureC* transcription, despite the presence of *ureC* genes in marine AOA natural populations, add to the complexity of the issue²⁴. In summary, existing gene expression and rate measurement-based field studies have not provided coherent and definitive conclusions regarding direct urea oxidation by marine AOA in marine environments. Additional studies are required to further explore urea oxidation activities in the open ocean and mesopelagic layers. In addition to the suggested competitive advantages of utilizing both ammonia and urea, we observed that *Nitrosopumilus*-like marine AOA can increase their affinity

for ammonia upon ammonia depletion (Fig. 3). This finding suggests an additional adaptive capacity that enables marine AOA to more efficiently scavenge ammonia released from the decomposition of urea or other organic matters by surrounding microorganisms.

Initial description of the ureolytic capacity of *Nitrosospira lacus*

It was previously shown via nitrite accumulation that *Nitrosospira lacus* is ureolytic²⁵. However, those experiments did not reveal any N substrate preference. Our comprehensive monitoring of all N species and ¹⁵N isotope labeling during growth revealed the unexpected preferential utilization of urea over ammonia in β -AOB (Figs. 1d and 2). This finding underscores the importance of nitrogen uptake regulation for nitrogen metabolism of different AOM lineages. In addition, we observed in the mixed media that the specific growth rates of *N. lacus* and *N. ureae* on urea were comparable or even higher than on ammonia (Fig. 1d and f). These results provide a mechanistic basis for the initial findings reported by Urakawa *et al*²⁵.

Potential adaptation of non-ureolytic β -AOB in the environment

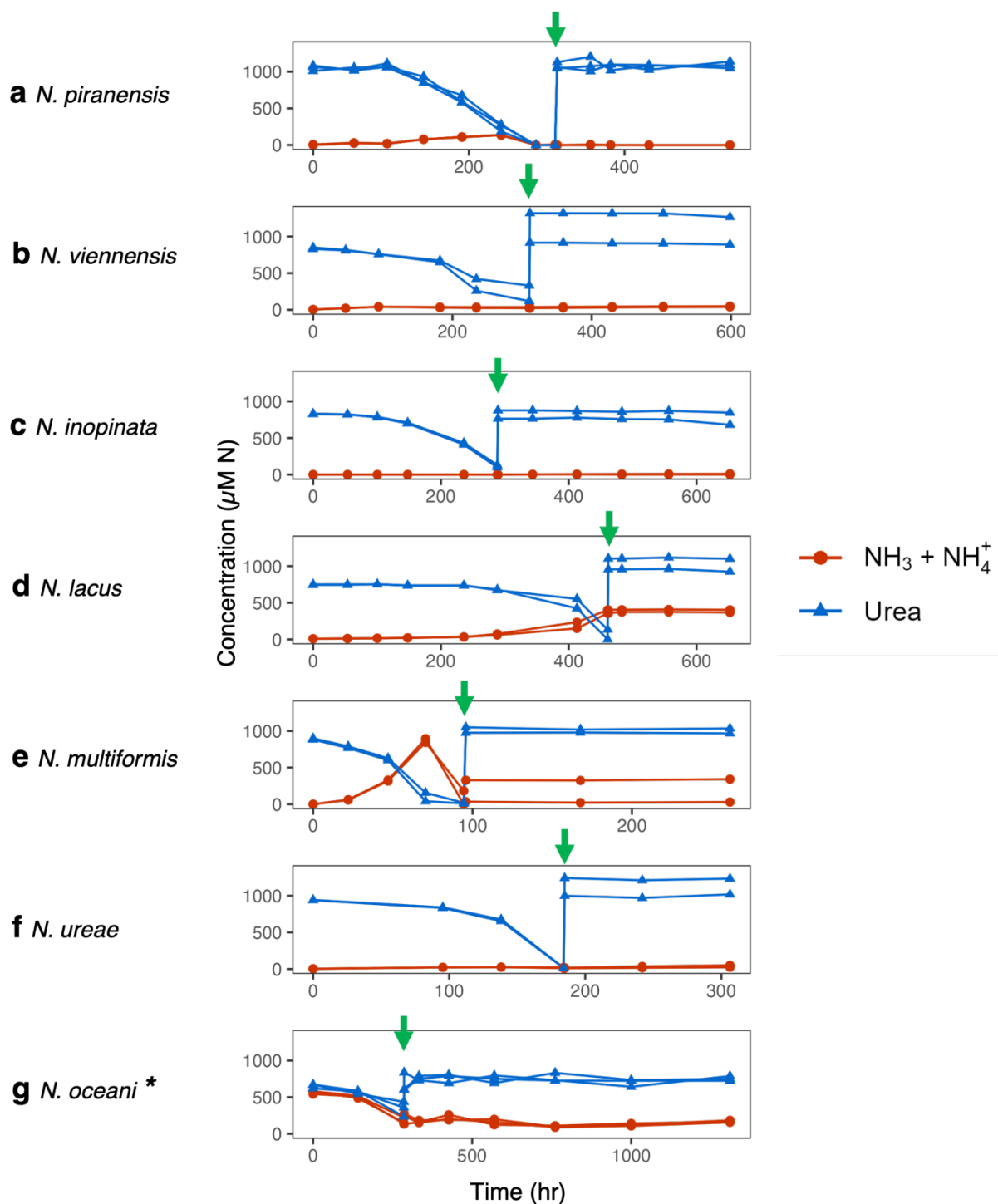
Our comparative genomics analysis reveals the widespread capacity for urea uptake and utilization among β -AOB (Extended Data Fig. 1b). The preference for urea utilization by β -AOB may serve as an adaptive strategy, potentially reducing direct competition with AOA and comammox for ammonia in ammonia-limited environments. In contrast, non-ureolytic β -AOB populations lacking urease genes might possess an advantage over other AOM species in transitioning in and out of dormancy in response to ammonia starvation, thereby maintaining microbial persistence in environments with fluctuating energy and nutrient availability^{26,27}.

Considerations and future research on N substrate selection among different AOM lineages

The AOM species characterized in our study exhibited a diverse range of adaptive strategies to coordinate growth on a mixture of ammonia and urea. We acknowledge the need for a wider range of ecotypes in AOM pure cultures to further investigate the diversity of N substrate

277 preferences and the regulatory systems among globally abundant ammonia oxidizers. For
278 example, it will be of interest to investigate *Nitrosopelagic*-like marine AOA species dominant in
279 the epipelagic ocean, the Water Column B (WCB) marine AOA dominant in the mesopelagic
280 ocean, *Nitrosotalea*-like soil AOA species dominant in acidic soils, *Nitrosocosmicus*-like AOA
281 species abundant in some high-ammonia environments, as well as clade B comammox species.
282 These future studies will offer a comprehensive understanding of the N substrate selection
283 regulation in AOM and provide valuable insights into the widely observed coexistence patterns
284 of different AOM lineages in various environments.

285

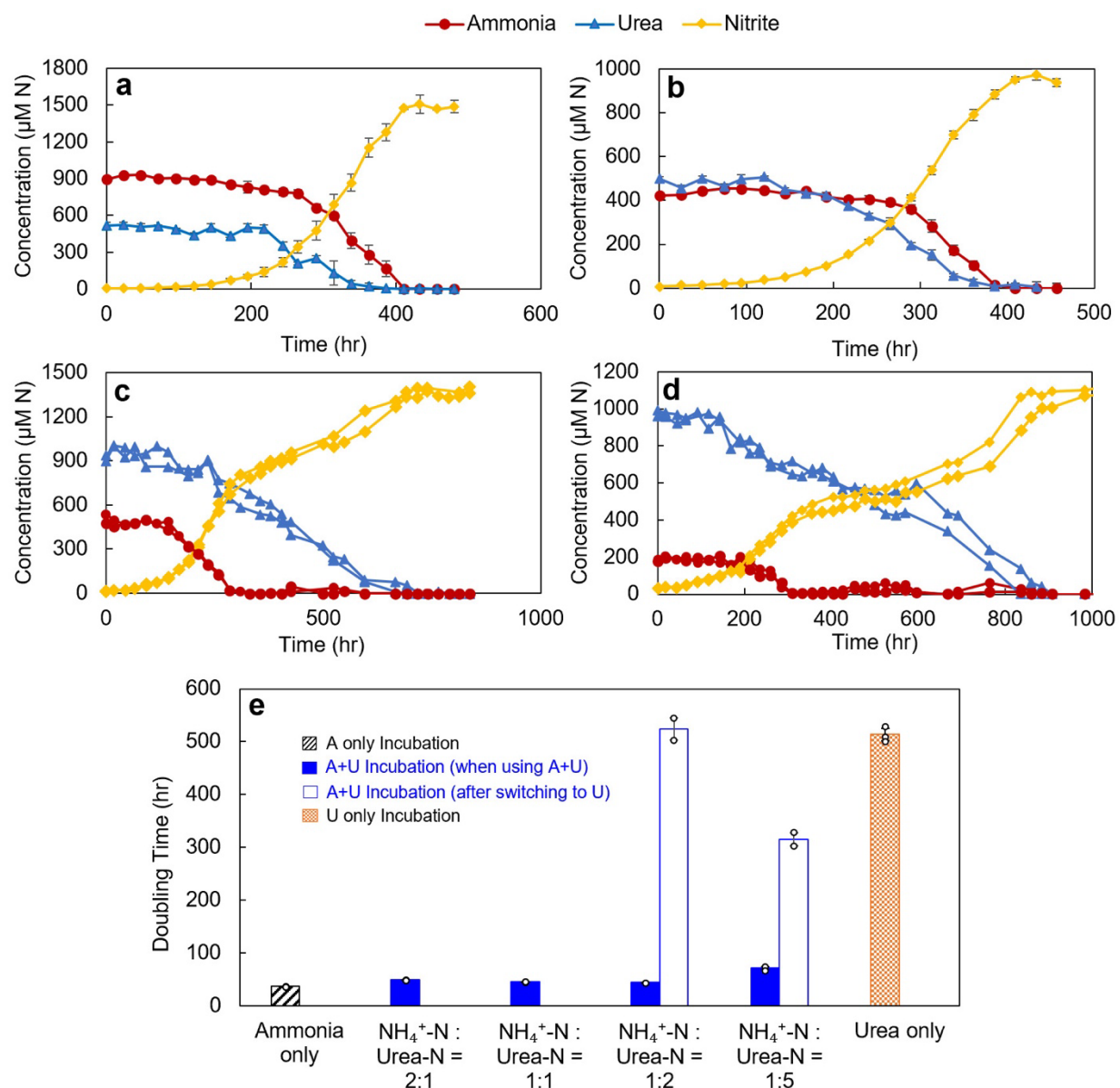


287

288 **Supplementary Figure 1. No extracellular ureolytic activity was found for marine and soil**

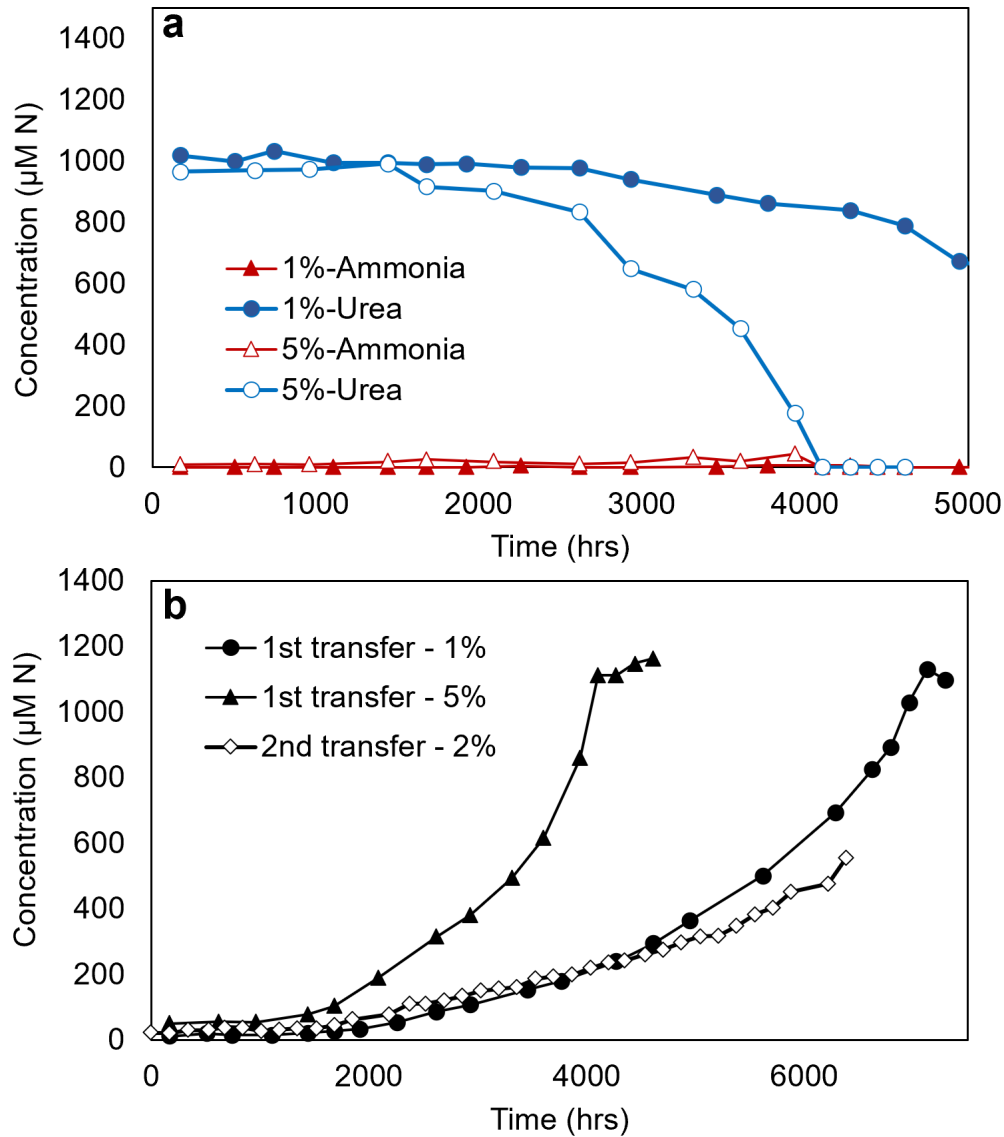
289 **AOA, β - and γ -AOB, and comammox species. Concentration profiles showing the cessation of**

290 urea utilization after removal of cells for **(a)** *N. piranensis* (marine AOA), **(b)** *N. viennensis* (soil
291 AOA), **(c)** *N. inopinata* (comammox), **(d)** *N. lacus* (*Nitrosospira* β -AOB), **(e)** *N. multiformis*
292 (*Nitrosospira* β -AOB), **(f)** *N. ureae* (*Nitrosomonas* β -AOB), and **(g)** *N. oceani* (γ -AOB). Lines
293 of the same shape/color represent biological replicates. Green arrow: time of cell removal (via
294 0.2 μ m filter) and re-addition of urea media. *Ammonia was supplied along with urea because
295 the presence of ammonia is needed for normal rate of urea consumption by *N. oceani*. Note that
296 neither were signal peptides identified in the urease genes of these characterized AOM species,
297 nor were secretion system genes found in the flanking regions of the urease operons, further
298 supporting the cytoplasmic localization of their urease enzymes.

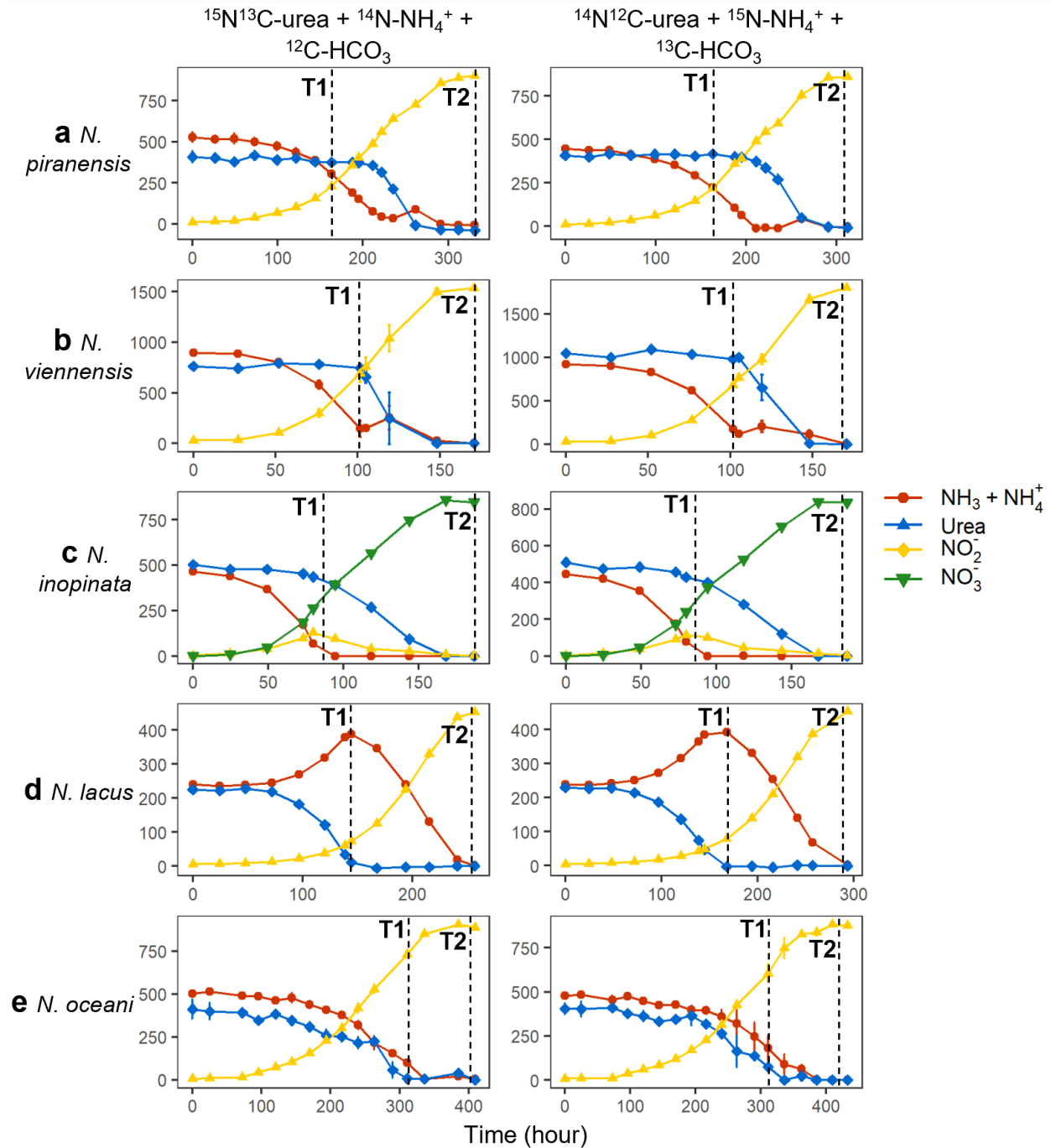


Supplementary Figure 2. Nitrogen source consumption of *Nitrosococcus oceanii* grown in medium with different concentrations of ammonia and urea. Incubation with ammonia-N:urea-N ratio of (a) 2:1, (b) 1:1, (c) 1:2, and (d) 1:5. e, Comparison of *N. oceanii* doubling time when grown in ammonia-only media, mixtures with different ammonia:urea ratios, and urea-only media. a and b, Error bars show the standard deviation of biological triplicates. c and d, Lines of the same shape/color represent biological duplicates. e, Note that some error bars are too

306 small to be visible, and error bars for the 1:2 and 1:5 incubations indicate the range of values
307 from biological duplicates.

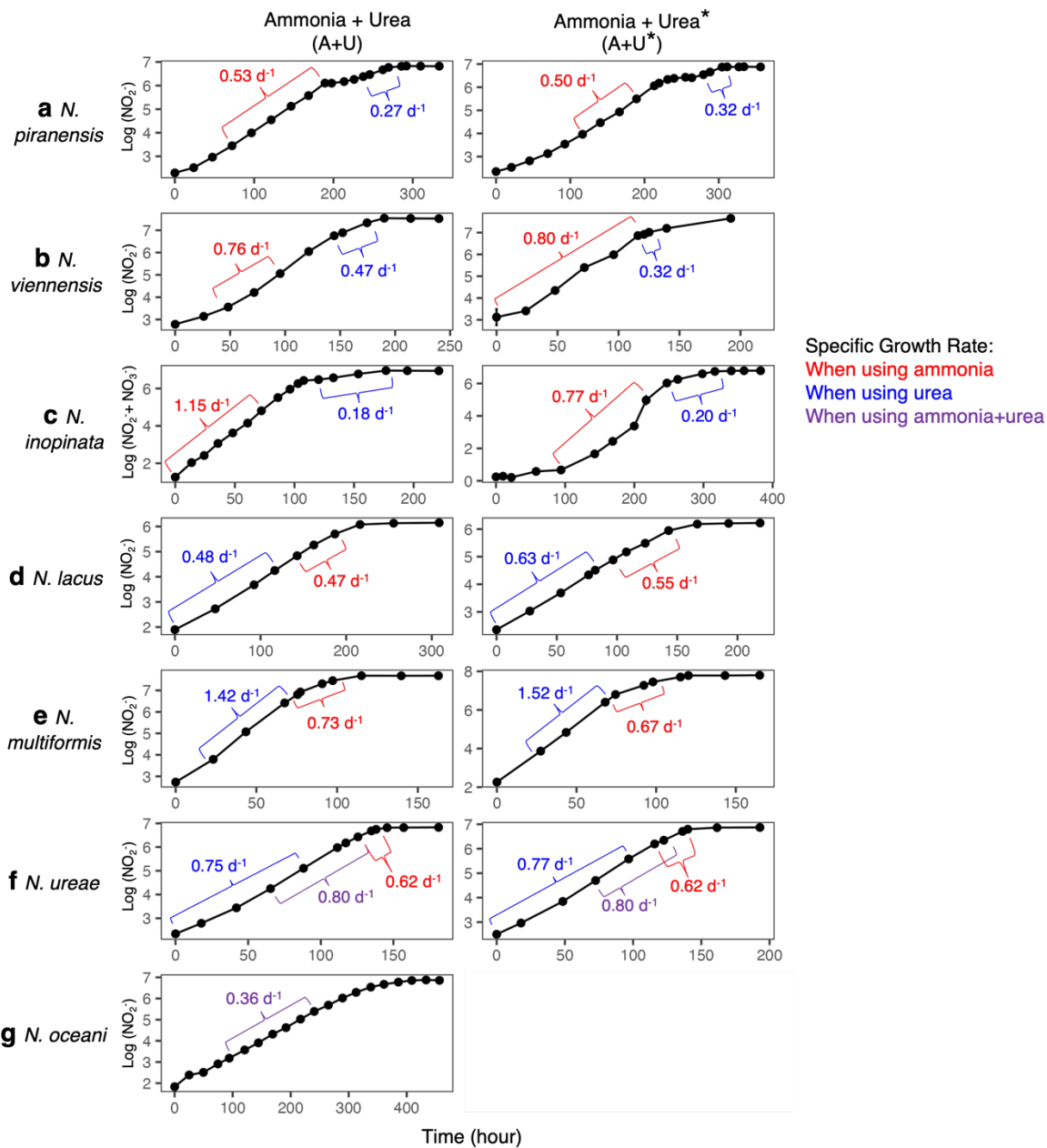


Supplementary Figure 3. Extremely slow urea consumption and nitrite production of *Nitrosococcus oceanii* in urea-only medium. Due to the slow growth rate of *N. oceanii* in urea-only medium, only two transfers were conducted. No appreciable increase in specific growth rate (0.03 d^{-1}) was found following the transfer of urea-grown *N. oceanii* into fresh urea-only medium. **a**, Concentrations of ammonia and urea of two incubations (1% and 5% inoculum) from the 1st transfer. **b**, Nitrite accumulation growth curves of the 1st and 2nd (2% inoculum) transfers.



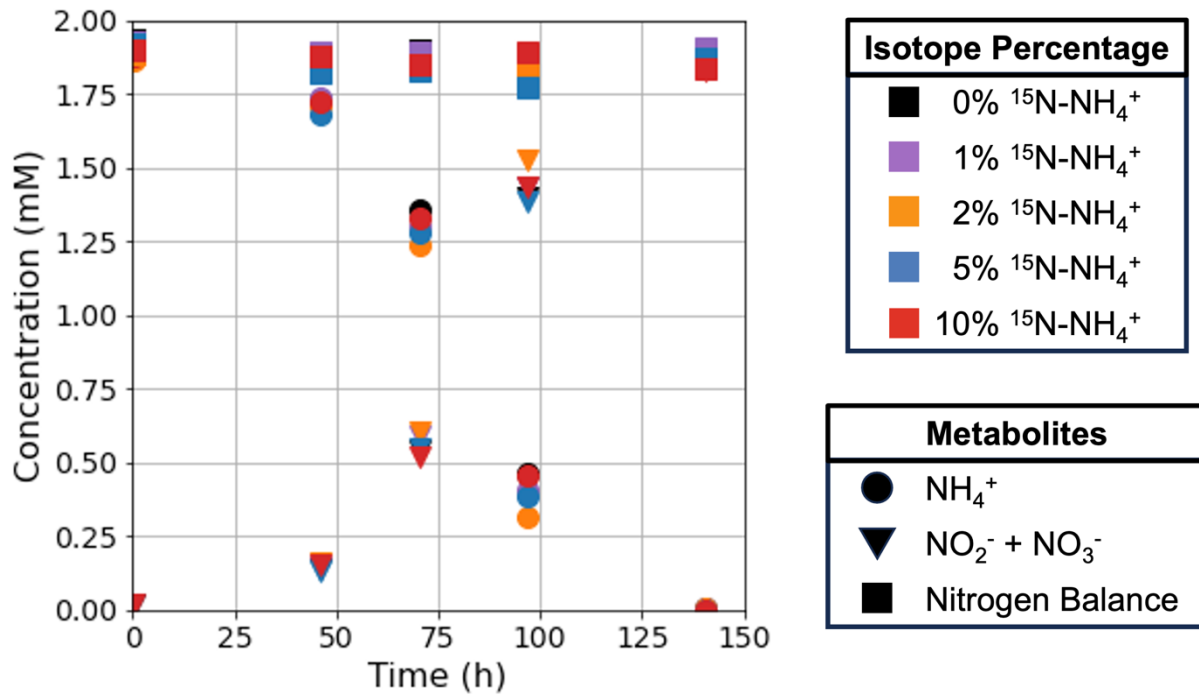
Supplementary Figure 4. Nitrogen nutrient data of incubations conducted for the NanoSIMS analysis. T1 and T2 indicate the two time points when cells were collected. (a) *N. piranensis* (marine AOA), (b) *N. viennensis* (soil AOA), (c) *N. inopinata* (comammox), (d) *N. lacus* (β -AOB), and (e) *N. oceani* (γ -AOB). Markers and error bars are average and one standard

321 deviation of biological triplicates except for the *N. viennensis* $^{15}\text{N-NH}_4^+ + ^{13}\text{C-HCO}_3$ incubation
322 (b, right), where error bars indicate the maximum and minimum values of two biological
323 replicates.

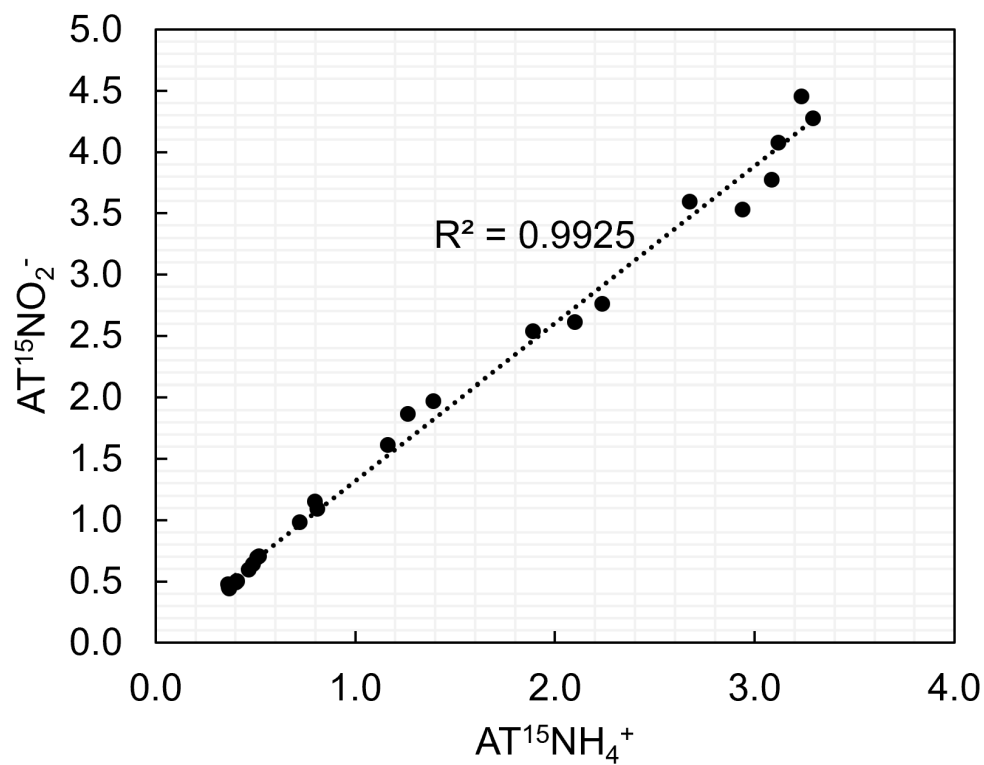


Supplementary Figure 5. Region of growth curves used to calculate the μ_{max} in Fig. 1 for the ammonia plus urea (A+U) incubations. (a) *N. piranensis* (marine AOA), (b) *N. viennensis* (soil AOA), (c) *N. inopinata* (comammox), (d) *N. lacus* (*Nitrosospira* β -AOB), (e) *N. multiformis* (*Nitrosospira* β -AOB), (f) *N. ureae* (*Nitrosomonas* β -AOB), and (g) *N. oceani* (γ -AOB). The A+U incubations were inoculated with 1% ammonia-grown culture whereas the A+U* incubations were inoculated with 1% urea-grown culture at mid-exponential phase.

331 Markers and error bars are average and one standard deviation of biological triplicates. The
332 standard deviation is smaller than the marker if error bar is not visible. Urea-only incubations
333 inoculated with urea-grown culture were not conducted for *N. oceanii* due to the extremely slow
334 growth rate.



Supplementary Figure 6. The ammonia oxidation activity of *N. inopinata* showed no discernible effect when varying ^{15}N substrate labeling percentages from 1 to 10%.



Supplementary Figure 7. Linear correlation between the newly produced $^{15}N\text{-}NO_2^-$ and released $^{15}N\text{-}NH_3$ by *Nitrosococcus oceani* in the growth experiments supplemented with $^{15}N\text{-}urea$. The linear regression line is $y = 1.284x + 0.0341$ ($R^2 = 0.9925$).

Supplementary Tables

Supplementary Table 1. *P*-values from two-tailed t-test ($\alpha = 0.05$) pairing the maximum specific growth rate between different incubation experiments. Values less than 0.05 are highlighted in green. Letters without square brackets indicate treatment with different N substrates: A = incubation with ammonia only; A+U = incubation with ammonia + urea; U = incubation with urea only. Letters in square brackets indicate the substrate being utilized when calculated for the μ_{max} : A = when using ammonia; U = when using urea; A+U = when using ammonia and urea together. *Inoculation with 1% urea-grown maintenance culture at mid-exponential phase; all other incubations were inoculated with 1% ammonia-grown maintenance culture.

Supplementary Table 2. The summarized ammonia- and urea-dependent oxidation kinetics parameters of *N. piranensis*, *N. viennensis*, *N. inopinata*, *N. lacus*, and *N. multiformis*.

Supplementary Table 3. The summarized transcription data of ammonia and urea uptake and utilization gene in AMO species.

Supplementary Table 4. Description of the time points collected for RNA sequencing for each tested AOM species.

AOM	Experiment	Time point 1	Time point 2	Time point 3	Time point 4
<i>N. piranensis</i>	AU	Exponential phase on ammonia	24 hr after urea spike	End of lag phase	Exponential phase on urea
	UA	Exponential phase on urea	24 hr after ammonia spike	Exponential phase on ammonia and urea	n/a
<i>N. viennensis</i>	AU	Exponential phase on ammonia	24 hr after urea spike	n/a	n/a
	UA	Exponential phase on urea	24 hr after ammonia spike	n/a	n/a
<i>N. inopinata</i>	AU	Exponential phase on ammonia	24 hr after urea spike	Exponential phase on urea	n/a
	UA	Exponential phase on urea	24 hr after ammonia spike	Right before ammonia depletion	n/a
<i>N. lacus</i>	A+U	Exponential phase on urea	8 hr before urea depletion	Exponential phase on ammonia	n/a
<i>N. oceani</i>	AU	Exponential phase on ammonia	24 hr after urea spike	Exponential phase on ammonia and urea	n/a
	UA	Not conducted due to extremely slow specific growth rate in urea-only medium.			

368 **Source data**

369 Source Data Table 1. Differential transcription of all AMO species genes in response to
370 ammonia and urea additions.

371 Source Data Table 2. NanoSIMS analysis output (“Raw”) and processed data (“Data”) for
372 calculating substrate incorporation percentages.

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